

CORE VARIETIES OF MOMENT MATRICES WITH LOW-DEGREE COLUMN RELATIONS

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ABSTRACT. By the Core Variety Theorem of Blekherman and Fialkow, the core variety of a truncated moment sequence is the union of the supports of all its representing measures; consequently, its nonemptiness characterizes the existence of a representing measure. Explicit computation of the core variety in several variables is generally difficult. We determine the possible core varieties for positive semidefinite bivariate moment matrices in several low-degree settings. For linear and quadratic relations, we treat the pure cases and recursively generated non-pure relations defined by a line, an irreducible conic with infinite real locus, or two distinct real lines. For cubic relations, we assume P -purity and consider P irreducible, the product of a real line and an irreducible conic with infinite real locus, or the product of three distinct real lines. The cubic classification depends on the irreducible decomposition of P and the geometry of the intersections of its components. The resulting core varieties are either the full real algebraic variety, the empty set, a finite extremal set, or, in certain reducible cases, a union of full components and finitely many points on the remaining component.

1. INTRODUCTION

Let $\beta \equiv \beta^{(m)} := \{\beta_{ij} : i, j \in \mathbb{Z}_+, i + j \leq m\}$ be a real bivariate sequence of order m , with $\beta_{00} > 0$. In the *truncated real moment problem* (TRMP), one asks whether there is a positive Borel measure μ on $\mathbb{R}^2$ such that $\int x^i y^j d\mu = \beta_{ij}$ for every $i + j \leq m$. Such a measure is called a *representing measure* for β . For $m = 2n$, the data are organized in the *moment matrix* $M(n) \equiv M(n)(\beta^{(2n)})$ as follows. Define the index set $\mathcal{A}_n = \{(i, j) \in \mathbb{N}_0^2 \mid i + j \leq n\}$ and endow it with the degree lexicographic order $\prec$; i.e., $(i, j) \prec (k, l)$ if either $i + j < k + l$, or both $i + j = k + l$ and $i > k$. Indexed by the totally ordered set $(\mathcal{A}_n, \prec)$, the entries of the moment matrix $M(n)$ are defined as:

$$M(n)_{(i,j), (k,l)} = \beta_{i+k, j+l}.$$

Let $\mathcal{P} := \mathbb{R}[x, y]$ and let $\mathcal{P}_n := \{q \in \mathcal{P} : \deg q \leq n\}$. The associated Riesz functional is $L_\beta : \mathcal{P}_{2n} \rightarrow \mathbb{R}$, defined by $L_\beta(\sum_{i+j \leq 2n} c_{ij} x^i y^j) := \sum_{i+j \leq 2n} c_{ij} \beta_{ij}$. For β to admit a representing measure, the moment matrix $M(n)$ must, at minimum, be positive semidefinite. In addition to this requirement, we consider properties related to the column space

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of $M(n)$, whose columns are indexed by monomials in $\mathcal{P}_n$ in degree-lexicographic order. For $p(x, y) = \sum_{i+j \leq n} a_{ij} x^i y^j \in \mathcal{P}_n$, we define $p(X, Y) := \sum_{i+j \leq n} a_{ij} X^i Y^j$, where $X^i Y^j$ denotes the column of $M(n)$ indexed by the monomial $x^i y^j$. This correspondence is referred to as the functional calculus associated with $M(n)$. A relation $p(X, Y) = \mathbf{0}$ is called a *column relation* of $M(n)$. Functional calculus allows us to interpret matrix dependencies as algebraic varieties, which determine the properties of the representing measure. A key prerequisite for the existence of such a measure is that $M(n)$ is *recursively generated*. Formally, this means the implication

$$p(X, Y) = \mathbf{0} \implies (pq)(X, Y) = \mathbf{0}$$

must hold for every polynomial q where $\deg q \leq n - \deg p$. Matrices failing this condition cannot admit a representing measure, making recursive generation a vital diagnostic tool in TRMP.

For a fixed column relation $p(X, Y) = \mathbf{0}$, we say that $M(n)$ is *p-pure* if its column-relation space is exactly $p\mathcal{P}_{n-\deg p}$; equivalently, $q(X, Y) = \mathbf{0}$ for $q \in \mathcal{P}_n$ if and only if $q = ph$ for some $h \in \mathcal{P}_{n-\deg p}$.

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and

$$\mathcal{Z}_{\mathbb{F}}(p) := \{(x, y) \in \mathbb{F}^2 : p(x, y) = 0\}$$

be the zero set of p over $\mathbb{F}$ and define the *algebraic variety* of β by

$$\mathcal{V} \equiv \mathcal{V}(M(n)) \equiv \mathcal{V}(\beta) := \bigcap_{p(X,Y)=\mathbf{0}, \deg \mathbf{p} \leq \mathbf{n}} \mathcal{Z}_{\mathbb{R}}(p). \quad (1.1)$$

As in [4], if $M(n)$ admits a representing measure μ , then

$$\text{supp } \mu \subseteq \mathcal{V}(\beta) \quad \text{and} \quad \text{rank } M(n) \leq \text{card } \text{supp } \mu \leq \text{card } \mathcal{V}(\beta).$$

In particular, $\text{rank } M(n) \leq \text{card } \mathcal{V}(\beta)$; this is the *variety condition*.

Set $\mathcal{CV}^{(0)}(\beta) := \mathcal{V}(\beta)$. For $i \geq 0$, define

$$\mathcal{N}_i := \{f \in \mathcal{P}_{2n} : L_{\beta}(f) = 0, f|_{\mathcal{CV}^{(i)}(\beta)} \geq 0\}$$

and

$$\mathcal{CV}^{(i+1)}(\beta) := \mathcal{CV}^{(i)}(\beta) \cap \bigcap_{f \in \mathcal{N}_i} \mathcal{Z}_{\mathbb{R}}(f).$$

The *core variety* of β (or of $M(n)(\beta)$), introduced by Fialkow in [13], is

$$\mathcal{CV}(\beta) \equiv \mathcal{CV}(M(n)) := \bigcap_{i=0}^{\infty} \mathcal{CV}^{(i)}(\beta).$$

The following summarizes the essential results needed to explicitly determine the core variety of truncated moment sequences.

Proposition 1 ([13, Proposition 2.1]). *If μ is a representing measure for β , then $\text{supp } \mu \subseteq \mathcal{CV}(\beta)$.*

Proposition 2 ([13, Proposition 2.3]). (i) *Every representing measure for β satisfies $\text{rank } M(n) \leq \text{card } \mathcal{CV}(\beta)$.*

(ii) *If a representing measure μ for β satisfies $\text{int}(\text{supp } \mu) \neq \emptyset$, then $\mathcal{CV}(\beta) = \mathbb{R}^2$.*

Note that Proposition 2(i) can be regarded as a *generalized variety condition*.

Proposition 3 ([13], Proposition 2.17). *Let $p(x, y) \in \mathbb{R}[x, y]$ be an irreducible polynomial with $\deg p = n$ and $\mathcal{Z}_{\mathbb{R}}(p)$ infinite. Suppose there exists a positive finite Borel measure μ on $\mathbb{R}^2$ with $\text{supp } \mu = \mathcal{Z}_{\mathbb{R}}(p)$. Then $\beta \equiv \beta^{(2n)}[\mu]$ satisfies $\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(p)$. Here $\beta^{(2n)}[\mu]$ denotes the sequence generated by a measure μ .*

Theorem 4 ([13, Theorem 2.20]). *If $\text{rank } M(n) = \text{card } \mathcal{CV}(\beta)$, then β admits a unique representing measure μ , and $\text{supp } \mu = \mathcal{CV}(\beta)$.*

Proposition 1 shows that if β has a representing measure, then $\mathcal{CV}(\beta)$ is nonempty. The main result concerning the core variety is the following converse, due to Blekherman and Fialkow.

Theorem 5 ([2, Core Variety Theorem]). *β has a representing measure if and only if $\mathcal{CV}(\beta)$ is nonempty. In this case, β has a finitely atomic representing measure, and the union of the supports of such measures is precisely $\mathcal{CV}(\beta)$.*

Recall Hilbert's theorem on sums of squares: a positive polynomial of degree $2n$ is always a sum of squares only in the univariate case, in degree 2, or in the bivariate quartic case. The truncated moment problems for these 3 cases have been solved and we also have:

Proposition 6 ([13, Proposition 3.1]). *In the cases of Hilbert's Theorem, if $M(n)$ is positive definite, then $\mathcal{CV}(\beta) = \mathbb{R}^2$ and β has a representing measure.*

An important solvable instance is the *extremal* case, in which $\text{rank } M(n) = \text{card } \mathcal{V}(\beta)$. In this case, a representing measure, if one exists, is unique (see [8, Theorem 1.3]). When an extremal moment sequence has a representing measure, its algebraic and core varieties coincide. Beyond the extremal case, the non-extremal sextic problems with $\text{rank } M(3) \in \{7, 8\}$ were settled in [10] by a rank-reduction technique originating in the solution of the nonsingular quartic problem [11]; here we use a related point-mass subtraction to treat the complementary case of a single cubic column relation, where $\text{rank } M(n) = 3n$ for $n \geq 3$, and we describe not only whether a representing measure exists but also all points that can occur as its atoms.

Main Results. In this paper we study the core variety of truncated moment sequences whose moment matrices admit column relations generated by algebraic curves of degree at most three. Our aim is to determine all possible forms of the core variety in terms of the geometry of the underlying algebraic curve.

The main classification results are summarized in Table 1. In the table, *full* means that $\mathcal{CV}(\beta) = \mathcal{V}(\beta)$, while *finite extremal* means that $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n)$, in which case the representing measure is unique and its support is $\mathcal{CV}(\beta)$.

Thus, the possible structure of the core variety is governed by the irreducible decomposition of the defining curve and by the intersection geometry of its components.

Organization of the paper. Section 2 collects the preliminary rank and perturbation results used throughout the paper. Section 3 treats linear and quadratic column relations, while Section 4 establishes the classification for irreducible cubic relations and provides

TABLE 1. Classification of the core varieties considered in this paper.

Defining relation	Hypotheses	Possible core varieties	Reference
Linear or quadratic p	p -pure	Full.	Theorem 10
Linear or irreducible quadratic p	Not p -pure; $M(n)$ recursively generated	Finite extremal, or empty.	Theorem 12
Reducible conic $p = p_1 p_2$	Not p -pure; p_1, p_2 nonproportional real linear polynomials	One full line C_i together with a set $S \subseteq C_j \setminus C_i$, where $\text{card } S \leq n - 1$; finite extremal; or empty.	Theorem 14
Irreducible cubic P	P -pure	Full; finite extremal of cardinality $3n$; or empty. The finite-extremal case is impossible when P is affinely equivalent to $y - x^3$.	Proposition 16; Theorem 17
Line C_1 and irreducible conic C_2	$P = P_1 P_2$ is P -pure; the components meet at two nonreal affine points	Full; $C_2 \cup S_1$, where $S_1 \subseteq C_1$ and $\text{card } S_1 = n - 1$; or empty.	Theorem 26
Line C_1 and irreducible conic C_2	$P = P_1 P_2$ is P -pure; every common affine zero is real; put $r := \text{card}(C_1 \cap C_2)$	Full; if $r \leq 1$, either $C_1 \cup S_2$ with $\text{card } S_2 = 2n - 1$, or $C_2 \cup S_1$ with $\text{card } S_1 = n - 1$; finite extremal of cardinality $3n$; or empty. If $r = 2$, the two intermediate cases are impossible.	Theorem 28
Three distinct real lines	$P = P_1 P_2 P_3$ is P -pure	Full; two full line components together with $n - 1$ points on the third line; finite extremal of cardinality $3n$; or empty. The intermediate case can occur only when the lines are concurrent or pairwise parallel.	Theorem 31

examples for the different affine types. Section 5 considers cubics that factor as a line times an irreducible conic, with particular attention to the intersection geometry of the two components. Section 6 treats cubics consisting of three real lines. The final section records two consequences for low-order positive definite moment matrices.

2. PRELIMINARIES

We begin by collecting notation and several elementary facts that will be used repeatedly in the sequel. Of particular importance are rank-one moment matrices associated with point masses and determinant formulas for their perturbations. These facts allow us to subtract a small multiple of a Dirac mass from a moment matrix while controlling positive

semidefiniteness, a technique that plays a central role in identifying points belonging to the core variety.

In this paper, we adopt the following conventions:

- (i) For $(a, b) \in \mathbb{R}^2$, let $\mathbf{v}_k(a, b)$ denote the row vector of monomials up to degree k in degree-lexicographic order:

$$\mathbf{v}_k(a, b) := (1 \quad a \quad b \quad a^2 \quad ab \quad b^2 \quad \cdots \quad b^k).$$

The matrix $P_k(a, b)$ is defined as the outer product

$$P_k(a, b) := \mathbf{v}_k(a, b)^T \mathbf{v}_k(a, b),$$

which represents the rank-one moment matrix of degree $2k$ associated with the Dirac measure $\delta(a, b)$.

- (ii) A_J signifies the compression of a square matrix A to the rows and columns corresponding to an index set J .
- (iii) Similarly, $M(n)_J$ represents the moment matrix $M(n)$ restricted to the rows and columns indexed by the subset of monomials J .
- (iv) For a real matrix A , let $\mathcal{C}(A) = \text{Ran } A$ denote its column space, and let $A^\dagger$ denote its Moore–Penrose pseudoinverse. For a symmetric block matrix

$$S = \begin{bmatrix} \mathbf{a} & \mathbf{b}^\dagger \\ \mathbf{b} & A \end{bmatrix},$$

we write

$$S/A := \mathbf{a} - \mathbf{b}^\dagger A^\dagger \mathbf{b}$$

for the generalized Schur complement of A in S . If A is invertible, then $S/A = \mathbf{a} - \mathbf{b}^\dagger A^{-1} \mathbf{b}$.

We shall also use the following standard rank-one downdating criterion.

Lemma 7. *Let $A \succeq 0$ and $\mathbf{u} \in \text{Ran } A \setminus \{\mathbf{0}\}$. If $0 < \rho < (\mathbf{u}^t A^\dagger \mathbf{u})^{-1}$, then $A - \rho \mathbf{u} \mathbf{u}^t \succeq 0$ and $\ker(A - \rho \mathbf{u} \mathbf{u}^t) = \ker A$.*

Proof. Write $\mathbf{u} = \sqrt{A} \mathbf{v}$, where $\mathbf{v} = \sqrt{A^\dagger} \mathbf{u}$. Then $A - \rho \mathbf{u} \mathbf{u}^t = \sqrt{A}(I - \rho \mathbf{v} \mathbf{v}^t) \sqrt{A}$. Since $\rho \|\mathbf{v}\|^2 = \rho \mathbf{u}^t A^\dagger \mathbf{u} < 1$, the middle factor is positive definite, which proves both assertions. $\square$

For $D \subseteq \mathbb{R}^2$, write

$$\mathcal{P}_n|_D := \{p|_D : p \in \mathcal{P}_n\} \quad \text{and} \quad \rho_n(D) := \dim(\mathcal{P}_n|_D).$$

The following elementary estimate controls the rank contributed by a curve component and finitely many additional points.

Lemma 8. *Let $n \geq 2$, let $D \subseteq \mathbb{R}^2$, and let μ be a positive finite Borel measure supported on $D \cup S$, where $S \subseteq \mathbb{R}^2$ is finite. Then*

$$\text{rank } M(n)[\mu] \leq \rho_n(D) + \text{card}(S \setminus D).$$

In particular, if C_1 is a line, C_2 is an irreducible conic with infinite real zero set, and L_1, L_2 are two distinct lines, then $\rho_n(C_1) = n + 1$, $\rho_n(C_2) = 2n + 1$, $\rho_n(L_1 \cup L_2) = 2n + 1$.

Proof. Consider the symmetric bilinear form $B_\mu : \mathcal{P}_n \times \mathcal{P}_n \rightarrow \mathbb{R}$ defined by $B_\mu(p, q) := \int pq d\mu$. Its matrix in the monomial basis of $\mathcal{P}_n$ is $M(n)[\mu]$.

If $p \in \mathcal{P}_n$ vanishes on $\text{supp } \mu$, then $B_\mu(p, q) = 0$ for every $q \in \mathcal{P}_n$. Conversely, suppose that $p \in \mathcal{P}_n$ satisfies $B_\mu(p, q) = 0$ for every $q \in \mathcal{P}_n$. Taking $q = p$ gives $\int p^2 d\mu = 0$. Since μ is positive and p is continuous, p vanishes on $\text{supp } \mu$. Indeed, if $p(z) \neq 0$ at some $z \in \text{supp } \mu$, there exist a neighborhood U of z and $\varepsilon > 0$ such that $p^2 \geq \varepsilon$ on U . By the definition of support, $\mu(U) > 0$, contradicting $\int p^2 d\mu = 0$. Thus the kernel of B_μ coincides with the kernel of restriction to $\text{supp } \mu$. Hence B_μ factors through the restriction space $\mathcal{P}_n|_{\text{supp } \mu}$, and therefore

$$\text{rank } M(n)[\mu] = \rho_n(\text{supp } \mu).$$

Since $\text{supp } \mu \subseteq D \cup S$, restriction first to D and then to the points in $S \setminus D$ gives

$$\rho_n(\text{supp } \mu) \leq \rho_n(D \cup S) \leq \rho_n(D) + \text{card}(S \setminus D).$$

This proves the rank estimate.

It remains to calculate the stated restriction dimensions. If $C_1 = \mathcal{Z}_{\mathbb{R}}(\ell)$, where ℓ is a nonzero linear polynomial, then the kernel of the restriction map $\mathcal{P}_n \rightarrow \mathcal{P}_n|_{C_1}$ is $\ell\mathcal{P}_{n-1}$. Consequently,

$$\rho_n(C_1) = \dim \mathcal{P}_n - \dim \mathcal{P}_{n-1} = \binom{n+2}{2} - \binom{n+1}{2} = n+1.$$

If $C_2 = \mathcal{Z}_{\mathbb{R}}(Q)$, where Q is irreducible and C_2 is infinite, then $I(C_2) = (Q)$, which denotes the principal ideal generated by Q . Thus the kernel of $\mathcal{P}_n \rightarrow \mathcal{P}_n|_{C_2}$ is $Q\mathcal{P}_{n-2}$, and

$$\rho_n(C_2) = \dim \mathcal{P}_n - \dim \mathcal{P}_{n-2} = \binom{n+2}{2} - \binom{n}{2} = 2n+1.$$

Finally, write $L_1 = \mathcal{Z}_{\mathbb{R}}(\ell_1)$ and $L_2 = \mathcal{Z}_{\mathbb{R}}(\ell_2)$, where ℓ_1, ℓ_2 are nonproportional linear polynomials. Since $I(L_1 \cup L_2) = (\ell_1 \ell_2)$, the kernel of the corresponding restriction map is $\ell_1 \ell_2 \mathcal{P}_{n-2}$. Therefore $\rho_n(L_1 \cup L_2) = \dim \mathcal{P}_n - \dim \mathcal{P}_{n-2} = 2n+1$. $\square$

3. THE CORE VARIETY $\mathcal{CV}(\beta)$ FOR MOMENT MATRICES $M(n)$ WITH A LINEAR OR QUADRATIC COLUMN RELATION

In this section, we characterize the possible cases for the core variety when the moment matrix $M(n)$ admits a linear or quadratic column relation.

We begin by recalling the following solution to the p -pure TMP for $p \in \mathbb{R}[x, y]_{\leq 2}$.

Theorem 9 ([5, 6, 7, 9, 12]). *Let $p \in \mathbb{R}[x, y]$ be a nonconstant polynomial with $\deg p \leq 2$. Suppose $M(n)(\beta^{(2n)})$ is positive semidefinite and p -pure. Then $\beta^{(2n)}$ admits a finitely atomic representing measure supported on $\mathcal{Z}_{\mathbb{R}}(p)$.*

Consequently, in the p -pure setting with $\deg p \leq 2$, the core variety coincides with the entire algebraic curve $\mathcal{Z}_{\mathbb{R}}(p)$:

Theorem 10. *Let $p \in \mathbb{R}[x, y]$ be a nonconstant polynomial with $\deg p \leq 2$. Suppose $M(n)(\beta)$ is positive semidefinite and p -pure. Then $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = \mathcal{Z}_{\mathbb{R}}(p)$.*

Proof. Fix $w \in \mathcal{Z}_{\mathbb{R}}(p)$, and let $v := \mathbf{v}_n(w)^t$. If $q(X, Y) = \mathbf{0}$, then $q = ph$ for some $h \in \mathcal{P}_{n-\deg p}$ by p -purity, and hence $q(w) = 0$. Thus $v \perp \ker M(n)$, so $v \in \text{Ran } M(n)$. Lemma 7 shows that, for all sufficiently small $\rho > 0$, $\widetilde{M} := M(n) - \rho P_n(w)$ is positive semidefinite and has the same kernel as $M(n)$. It is therefore again a p -pure moment matrix. By Theorem 9, the corresponding sequence has a finitely atomic representing measure $\tilde{\mu}$ supported on $\mathcal{Z}_{\mathbb{R}}(p)$. Consequently, $\tilde{\mu} + \rho \delta_w$ represents β and contains w in its support.

Since $w \in \mathcal{Z}_{\mathbb{R}}(p)$ was arbitrary, the Core Variety Theorem gives $\mathcal{Z}_{\mathbb{R}}(p) \subseteq \mathcal{CV}(\beta)$. The reverse inclusion follows from $p(X, Y) = \mathbf{0}$, while p -purity gives $\mathcal{V}(\beta) = \mathcal{Z}_{\mathbb{R}}(p)$. Hence $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = \mathcal{Z}_{\mathbb{R}}(p)$. $\square$

We first record an interpolation fact for an irreducible curve of degree at most two.

Lemma 11. *Let $n \geq 2$, let $p \in \mathbb{R}[x, y]$ be either linear or an irreducible quadratic polynomial with infinite real zero set, put $d := \deg p$, and set $D := \mathcal{Z}_{\mathbb{R}}(p)$. If $E \subseteq D$ is finite and $\text{card } E \leq dn + 1$, then the restriction map $\mathcal{P}_n \rightarrow \mathbb{R}^E$ is surjective. In particular, $\rho_n(E) = \text{card } E$.*

Proof. Extend E , if necessary, to a set $E' \subseteq D$ of $dn + 1$ distinct points. If $q \in \mathcal{P}_n$ vanishes on E' , then $p \mid q$: for $d = 1$, this follows by restricting q to the line D , and for $d = 2$, it follows from Bézout's theorem because $2n + 1 > 2n$. Thus evaluation on E' is injective on $\mathcal{P}_n|_D$. Since $\dim(\mathcal{P}_n|_D) = dn + 1 = \text{card } E'$, it is an isomorphism. Projection from $\mathbb{R}^{E'}$ onto $\mathbb{R}^E$ proves the assertion. $\square$

The next theorem treats the non-pure case for an irreducible linear or quadratic relation. The hypothesis that the real zero set is infinite excludes the exceptional irreducible quadratics whose real loci are empty or consist of a single point.

Theorem 12. *Let $n \geq 2$, and let $p \in \mathbb{R}[x, y]$ be either linear or an irreducible quadratic polynomial with infinite real zero set. Suppose that $M(n) \equiv M(n)(\beta)$ is positive semidefinite, recursively generated, satisfies $p(X, Y) = \mathbf{0}$, and is not p -pure. Then exactly one of the following alternatives holds:*

- (i) $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n)$, $\mathcal{CV}(\beta) = \mathcal{V}(\beta)$, and there exists a unique representing measure whose support is $\mathcal{CV}(\beta)$;
- (ii) $\mathcal{CV}(\beta) = \emptyset$, and no representing measure exists.

Proof. Put $d := \deg p$, $D := \mathcal{Z}_{\mathbb{R}}(p)$, $\mathcal{V} := \mathcal{V}(\beta)$, and $\mathcal{K}_n := \{q \in \mathcal{P}_n : q(X, Y) = \mathbf{0}\}$. Recursive generation gives $p\mathcal{P}_{n-d} \subseteq \mathcal{K}_n$. Since $M(n)$ is not p -pure, there is $q \in \mathcal{K}_n$ with $p \nmid q$. Therefore $\mathcal{V} \subseteq D \cap \mathcal{Z}_{\mathbb{R}}(q)$, so Bézout's theorem gives $\text{card } \mathcal{V} \leq dn$.

If $\mathcal{CV}(\beta) = \emptyset$, the Core Variety Theorem gives alternative (ii). Hence assume that $\mathcal{CV}(\beta) \neq \emptyset$. By the same theorem, there exists a finitely atomic representing measure μ such that $\text{supp } \mu \subseteq \mathcal{CV}(\beta) \subseteq \mathcal{V}$. Set $r := \text{rank } M(n)$.

Consider the restriction map

$$R_{\mathcal{V}} : \mathcal{P}_n \longrightarrow \mathcal{P}_n|_{\mathcal{V}}, \quad q \longmapsto q|_{\mathcal{V}}.$$

By the definition of $\mathcal{V}$, every polynomial in $\mathcal{K}_n$ vanishes on $\mathcal{V}$. Thus $\mathcal{K}_n \subseteq \ker R_{\mathcal{V}}$, and $R_{\mathcal{V}}$ descends to a surjective linear map

$$\overline{R}_{\mathcal{V}} : \mathcal{P}_n / \mathcal{K}_n \longrightarrow \mathcal{P}_n|_{\mathcal{V}}.$$

Since $\mathcal{K}_n$ is the column kernel of $M(n)$, it follows that

$$\rho_n(\mathcal{V}) \leq \dim(\mathcal{P}_n / \mathcal{K}_n) = \text{rank } M(n) = r.$$

We have already proved that $\text{card } \mathcal{V} \leq dn$. Hence Lemma 11 gives $\rho_n(\mathcal{V}) = \text{card } \mathcal{V}$. Moreover, $M(n)[\mu] = M(n)$, because μ represents β , and $\text{supp } \mu \subseteq \mathcal{V}$. Therefore

$$\text{card } \mathcal{V} = \rho_n(\mathcal{V}) \leq r = \text{rank } M(n)[\mu] \leq \text{card}(\text{supp } \mu) \leq \text{card } \mathcal{V}.$$

All inequalities are consequently equalities. In particular,

$$r = \text{card } \mathcal{V} \quad \text{and} \quad \text{supp } \mu = \mathcal{V}.$$

For $w \in \mathcal{V}$, choose $\ell_w \in \mathcal{P}_n$ with $\ell_w(w) = 1$ and $\ell_w \equiv 0$ on $\mathcal{V} \setminus \{w\}$. Every representing measure ν is supported on $\mathcal{V}$, and therefore $\nu(\{w\}) = L_{\beta}(\ell_w^2)$. Thus all its masses are uniquely determined, so μ is the unique representing measure. The Core Variety Theorem now yields $\mathcal{CV}(\beta) = \mathcal{V}$, proving alternative (i). $\square$

For a reducible conic, an additional possibility occurs: the core variety may contain one entire line together with finitely many points on the other line. We shall use the following two-line interpolation lemma.

Lemma 13. *Let $n \geq 1$, and let $C_1, C_2 \subseteq \mathbb{R}^2$ be two distinct affine lines.*

- (i) *If $E \subseteq C_1 \cup C_2$ is finite and $\text{card}(E \cap C_i) \leq n$ for $i = 1, 2$, then the restriction map*

$$\mathcal{P}_n \longrightarrow \mathbb{R}^E, \quad f \longmapsto (f(z))_{z \in E},$$

is surjective. Consequently, $\rho_n(E) = \text{card } E$.

- (ii) *If $S \subseteq C_2 \setminus C_1$ is finite and $\text{card } S \leq n$, then*

$$\rho_n(C_1 \cup S) = n + 1 + \text{card } S.$$

Proof. Both assertions are invariant under invertible affine changes of variables.

To prove (i), suppose first that the lines intersect. We may assume that $C_1 = \{(x, y) : y = 0\}$ and $C_2 = \{(x, y) : x = 0\}$. Prescribe arbitrary values on E . If $(0, 0) \in E$, let c be its prescribed value and choose univariate polynomials u, v interpolating the prescribed values on $E \cap C_1$ and $E \cap C_2$, respectively, with $u(0) = v(0) = c$. Since each of these sets has at most n points, we may take $\deg u, \deg v \leq n - 1$. Then $f(x, y) := u(x) + v(y) - c$ has degree at most n and assumes all the prescribed values.

If $(0, 0) \notin E$, additionally prescribe $u(0) = v(0) = 0$. Each interpolation problem then has at most $n + 1$ nodes, so $\deg u, \deg v \leq n$, and $f(x, y) := u(x) + v(y)$ has the required values.

If the lines are parallel, we may assume that $C_1 = \{(x, y) : y = 0\}$ and $C_2 = \{(x, y) : y = 1\}$. Choose univariate polynomials u_1, u_2 of degree at most $n - 1$ interpolating the prescribed values on $E \cap C_1$ and $E \cap C_2$, respectively. Then

$$f(x, y) := (1 - y)u_1(x) + yu_2(x)$$

has degree at most n and realizes all the prescribed values. This proves (i).

For (ii), put $k := \text{card } S$. The upper bound $\rho_n(C_1 \cup S) \leq n+1+k$ is immediate. Let ℓ_1 be a linear polynomial defining C_1 , and write $S = \{z_1, \dots, z_k\}$. If $k > 0$, choose polynomials $h_1, \dots, h_k$ whose restrictions to C_2 have degree at most $k-1$ and satisfy

$$h_j(z_m) = \frac{\delta_{jm}}{\ell_1(z_m)}.$$

Then $r_j := \ell_1 h_j$ has degree at most $k \leq n$, vanishes on C_1 , and satisfies $r_j(z_m) = \delta_{jm}$. Thus the restrictions of $r_1, \dots, r_k$ contribute k independent directions in addition to the $n+1$ -dimensional space $\mathcal{P}_n|_{C_1}$. Hence $\rho_n(C_1 \cup S) \geq n+1+k$, proving equality. $\square$

Theorem 14. *Let $n \geq 2$, let $p = p_1 p_2$, where $p_1, p_2 \in \mathbb{R}[x, y]$ are nonproportional linear polynomials, and set $C_i := \mathcal{Z}_{\mathbb{R}}(p_i)$. Suppose that $M(n) \equiv M(n)(\beta)$ is positive semidefinite, recursively generated, satisfies $p(X, Y) = \mathbf{0}$, and is not p -pure. Then exactly one of the following alternatives holds:*

- (i) *There exist $\{i_1, i_2\} = \{1, 2\}$ and a finite set $S \subseteq C_{i_2} \setminus C_{i_1}$, with $\text{card } S \leq n-1$, such that $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = C_{i_1} \cup S$; in this case, β has a representing measure.*
- (ii) *$\text{card } \mathcal{CV}(\beta) = \text{rank } M(n)$, $\mathcal{CV}(\beta) = \mathcal{V}(\beta)$, and there exists a unique representing measure whose support coincides with $\mathcal{CV}(\beta)$.*
- (iii) *$\mathcal{CV}(\beta) = \emptyset$, and no representing measure exists.*

Proof. Put $C := C_1 \cup C_2$, $\mathcal{V} := \mathcal{V}(\beta)$, and $\mathcal{K}_n := \{q \in \mathcal{P}_n : q(X, Y) = \mathbf{0}\}$. Since $M(n)$ is recursively generated and $p(X, Y) = \mathbf{0}$, $p\mathcal{P}_{n-2} \subseteq \mathcal{K}_n$. This inclusion is strict because $M(n)$ is not p -pure. Hence there exists $q \in \mathcal{K}_n$ such that $p \nmid q$. Since $p, q \in \mathcal{K}_n$,

$$\mathcal{V} \subseteq C \cap \mathcal{Z}_{\mathbb{R}}(q).$$

If $\mathcal{CV}(\beta) = \emptyset$, the Core Variety Theorem gives alternative (iii). Hence assume that $\mathcal{CV}(\beta) \neq \emptyset$. By the same theorem, there exists a finitely atomic representing measure μ such that $\text{supp } \mu \subseteq \mathcal{CV}(\beta) \subseteq \mathcal{V}$. Set $r := \text{rank } M(n)$.

Every element of $\mathcal{K}_n$ vanishes on $\mathcal{V}$. Therefore restriction to $\mathcal{V}$ induces a surjective map

$$\mathcal{P}_n / \mathcal{K}_n \longrightarrow \mathcal{P}_n|_{\mathcal{V}},$$

and consequently

$$\rho_n(\mathcal{V}) \leq \dim(\mathcal{P}_n / \mathcal{K}_n) = r.$$

Suppose first that $\mathcal{V}$ is infinite. Then $\mathcal{V} \cap C_i$ is infinite for some $i \in \{1, 2\}$. Every polynomial in $\mathcal{K}_n$ therefore vanishes identically on C_i , and hence is divisible by p_i . It follows that $C_i \subseteq \mathcal{V}$ and, in particular, $q = p_i h$ for some $h \in \mathcal{P}_{n-1}$. Since $p \nmid q$, we have $p_{3-i} \nmid h$. Thus

$$\mathcal{V} = C_i \cup S, \quad S := \mathcal{V} \setminus C_i \subseteq C_{3-i} \setminus C_i,$$

where $k := \text{card } S \leq n-1$.

By Lemma 13, $\rho_n(\mathcal{V}) = n + 1 + k$. Write $\mu = \mu_i + \mu_S$, where $\mu_i := \mu|_{C_i}$ and $\mu_S := \mu|_S$. Then

$$\begin{aligned} n + 1 + k = \rho_n(\mathcal{V}) &\leq r = \text{rank } M(n)[\mu] \\ &\leq \text{rank } M(n)[\mu_i] + \text{rank } M(n)[\mu_S] \\ &\leq n + 1 + \text{card}(\text{supp } \mu_S) \leq n + 1 + k. \end{aligned}$$

Hence equality holds throughout. In particular,

$$\text{rank } M(n)[\mu_i] = n + 1, \quad \text{supp } \mu_S = S.$$

Since μ_i is supported on C_i , the kernel of $M(n)[\mu_i]$ contains $p_i \mathcal{P}_{n-1}$. Its rank is $n + 1 = \rho_n(C_i)$, so equality of dimensions shows that this is the entire kernel. Thus $M(n)[\mu_i]$ is p_i -pure. By Theorem 10, the core variety of the moment sequence generated by μ_i is C_i .

For every $z \in C_i$, there is therefore a representing measure $\nu_{i,z}$ for the moments of μ_i whose support contains z . Since $\nu_{i,z} + \mu_S$ represents β , we obtain $C_i \subseteq \mathcal{CV}(\beta)$. Moreover, $S = \text{supp } \mu_S \subseteq \text{supp } \mu \subseteq \mathcal{CV}(\beta)$. Consequently,

$$\mathcal{CV}(\beta) = \mathcal{V} = C_i \cup S,$$

which is alternative (i).

Suppose now that $\mathcal{V}$ is finite. For each $i \in \{1, 2\}$, there exists $q_i \in \mathcal{K}_n$ not divisible by p_i ; otherwise every element of $\mathcal{K}_n$ would vanish identically on C_i , giving $C_i \subseteq \mathcal{V}$, contrary to the finiteness of $\mathcal{V}$. Since $q_i|_{C_i}$ is a nonzero univariate polynomial of degree at most n ,

$$\text{card}(\mathcal{V} \cap C_i) \leq n, \quad i = 1, 2.$$

Lemma 13 therefore gives $\rho_n(\mathcal{V}) = \text{card } \mathcal{V}$. Hence

$$\text{card } \mathcal{V} = \rho_n(\mathcal{V}) \leq r = \text{rank } M(n)[\mu] \leq \text{card}(\text{supp } \mu) \leq \text{card } \mathcal{V}.$$

All inequalities are equalities. Thus

$$r = \text{card } \mathcal{V}, \quad \text{supp } \mu = \mathcal{V}, \quad \mathcal{CV}(\beta) = \mathcal{V}.$$

It remains to prove uniqueness. For every $w \in \mathcal{V}$, Lemma 13 provides $\ell_w \in \mathcal{P}_n$ such that $\ell_w(w) = 1$ and $\ell_w \equiv 0$ on $\mathcal{V} \setminus \{w\}$. If ν is any representing measure for β , then $\text{supp } \nu \subseteq \mathcal{V}$, and therefore

$$\nu(\{w\}) = \int \ell_w^2 d\nu = L_\beta(\ell_w^2).$$

Thus the mass of ν at every point of $\mathcal{V}$ is uniquely determined. Hence μ is the unique representing measure, proving alternative (ii). $\square$

Corollary 15. *Let $n \geq 2$, and suppose that $M(n)(\beta)$ has a representing measure. Suppose that $M(n)(\beta)$ admits a column relation $p(X, Y) = \mathbf{0}$, where p is linear, an irreducible quadratic with infinite real zero set, or a product of two nonproportional real linear polynomials. Then $\mathcal{CV}(\beta) = \mathcal{V}(\beta)$.*

Proof. The statement follows from Theorems 10, 12, and 14. $\square$

Corollary 15 suggests the following question:

Question. If $M(n)(\beta)$ admits a representing measure, does it follow that $\mathcal{CV}(\beta) = \mathcal{V}(\beta)$?

The answer is negative, already in the case of cubic curves (see Section 4).

4. THE CORE VARIETY $\mathcal{CV}(\beta)$ FOR MOMENT MATRICES $M(n)$ WITH AN IRREDUCIBLE CUBIC RELATION

In this section, we characterize the possible cases for the core variety when the moment matrix $M(n)$ admits an irreducible cubic column relation.

We begin by classifying irreducible planar cubic curves, up to invertible affine linear changes of variables, into thirteen normal forms.

Proposition 16 (Irreducible cubic normal forms). *Let $P \in \mathbb{R}[x, y]$ be an irreducible cubic polynomial. Up to an invertible affine linear change of variables, P is equivalent to one of the following real normal forms below. All parameters appearing below are real, and each parameterized family is understood to include only those values for which the displayed cubic is irreducible. Thus, the conditions displayed alongside the normal forms are not intended to constitute complete irreducibility criteria.*

(Weierstraß) *Smooth Weierstraß:* $y^2 - x(x - a)(x - b)$, $0 < a < b$, or
 $y^2 - x(x^2 + c^2)$, $c \neq 0$;

Neile's parabola: $y^2 - x^3$;

Nodal cubic: $y^2 - x(x - 1)^2$;

Isolated point: $y^2 - x^2(x - 1)$.

(Newton) *Rational cubic type 1:* $xy^2 + ay - dx - e$, $e^2 \neq a^2d$;

Non-Weierstraß type 1: $xy^2 + ay - x^2 - dx - e$;

Non-Weierstraß type 2: $xy^2 - x^3 - cx^2 - dx - e$, $e \neq 0$;

Non-Weierstraß type 3: $xy^2 + x^3 - cx^2 - dx - e$, $e \neq 0$;

Non-Weierstraß type 4: $xy^2 + ay - x^3 - cx^2 - dx - e$, $ae \neq 0$;

Non-Weierstraß type 5: $xy^2 + ay + x^3 - cx^2 - dx - e$, $ae \neq 0$.

(Rational) *Rational cubic type 2:* $xy - x^3 - q_2x^2 - q_1x - q_0$, $q_0 \neq 0$;

Cubical parabola: $y - x^3$.

Proof. This is the real affine classification of irreducible plane cubics in the form used in [15]; see in particular the case-by-case affine reductions there. The Weierstraß reductions may also be found in [1, Chapter III]. The displayed restrictions exclude certain immediate degeneracies; all parameter values remain subject to the irreducibility hypothesis of the proposition. $\square$

Recall from Section 3 that for lines and conics, pureness implies that the core variety coincides with the entire underlying curve. We now investigate the analogous phenomenon for cubic curves.

We extend the trichotomy of [13, Theorem 3.4], which classifies positive definite moment matrices $M(3)(\beta)$ according to whether $\mathcal{CV}(\beta) = \mathbb{R}^2$, $\text{card } \mathcal{CV}(\beta) = 10$, or $\mathcal{CV}(\beta) = \emptyset$, to the setting in which $\mathcal{CV}(\beta)$ is contained in an irreducible cubic curve. We also identify the cubical parabola as an exceptional case in which the finite-core alternative cannot occur.

Theorem 17. *Let $n \geq 3$, $P(x, y) \in \mathbb{R}[x, y]$ be an irreducible cubic polynomial, and let $M(n)(\beta)$ be a positive semidefinite moment matrix that is P -pure. Then exactly one of the following alternatives holds:*

- (i) $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = \mathcal{Z}_{\mathbb{R}}(P)$, and there exists a representing measure;
- (ii) $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n)(\beta) = 3n$, and there exists a unique representing measure whose support coincides with $\mathcal{CV}(\beta)$;
- (iii) $\mathcal{CV}(\beta) = \emptyset$, and no representing measure exists.

If P is affinely equivalent to $y - x^3$, alternative (ii) cannot occur, so only alternatives (i) and (iii) are possible.

Before proving Theorem 17, we record a general reduction concerning the first core iteration. The first assertion of the following lemma is an elementary finite-dimensional reduction of the first core iteration defined in [2]. The second assertion is a curve-specific consequence of irreducibility and Bézout's theorem.

Lemma 18. *Let $\beta \equiv \beta^{(2n)}$, set $C := \mathcal{V}(\beta)$, and define*

$$\mathcal{A}_0 := \{g \in \mathbb{R}[x, y]_{\leq 2n} : g|_C \geq 0, \quad L_{\beta}(g) = 0\}.$$

Then the following statements hold.

- (i) *There exists $F \in \mathcal{A}_0$ such that $\mathcal{CV}^{(1)}(\beta) = C \cap \mathcal{Z}_{\mathbb{R}}(F)$.*
- (ii) *Suppose, in addition, that $C = \mathcal{Z}_{\mathbb{R}}(P)$, where $P \in \mathbb{R}[x, y]$ is irreducible and C is infinite. If $\mathcal{CV}(\beta)$ is finite, then $\mathcal{CV}^{(1)}(\beta)$ is finite, and the polynomial F in (i) can be chosen so that $F \notin (P)$.*

Proof. Let $W := \text{span } \mathcal{A}_0$. Choose $f_1, \dots, f_r \in \mathcal{A}_0$ forming a basis of W ; if $W = \{0\}$, take $r = 1$ and $f_1 = 0$. By definition, $\mathcal{CV}^{(0)}(\beta) = \mathcal{V}(\beta)$ and

$$\mathcal{CV}^{(1)}(\beta) = C \cap \bigcap_{g \in \mathcal{A}_0} \mathcal{Z}_{\mathbb{R}}(g) = C \cap \bigcap_{j=1}^r \mathcal{Z}_{\mathbb{R}}(f_j).$$

Set $F := f_1 + \dots + f_r$. Since every f_j is nonnegative on C , for $z \in C$,

$$F(z) = 0 \iff f_1(z) = \dots = f_r(z) = 0.$$

Moreover, $F|_C \geq 0$, $L_{\beta}(F) = 0$, and $\deg F \leq 2n$. Hence $F \in \mathcal{A}_0$ and $\mathcal{CV}^{(1)}(\beta) = C \cap \mathcal{Z}_{\mathbb{R}}(F)$, which proves (i).

For (ii), suppose that $\mathcal{CV}(\beta)$ is finite but $\mathcal{CV}^{(1)}(\beta)$ is infinite. For every $g \in \mathcal{A}_0$, the intersection $C \cap \mathcal{Z}_{\mathbb{R}}(g)$ contains the infinite set $\mathcal{CV}^{(1)}(\beta)$. Since P is irreducible, Bézout's

theorem implies that $P \mid g$. Thus every element of $\mathcal{A}_0$ vanishes identically on C , and consequently $\mathcal{CV}^{(1)}(\beta) = C$.

The first core iteration has therefore produced no reduction. Accordingly, the defining family remains unchanged at every later iteration, so $\mathcal{CV}^{(j)}(\beta) = C$ for every $j \geq 1$. This gives $\mathcal{CV}(\beta) = C$, contradicting the assumptions that $\mathcal{CV}(\beta)$ is finite and C is infinite. Hence $\mathcal{CV}^{(1)}(\beta)$ is finite.

Finally, let F be the polynomial obtained in (i). If $F \in (P)$, then F vanishes identically on C , and hence $\mathcal{CV}^{(1)}(\beta) = C \cap \mathcal{Z}_{\mathbb{R}}(F) = C$, contradicting the finiteness of $\mathcal{CV}^{(1)}(\beta)$. Therefore $F \notin (P)$. $\square$

The preceding lemma reduces the first core iteration to the zero set of a single non-negative annihilating polynomial. To estimate the number of its zeros on an irreducible curve, we combine Bézout's theorem with the following parity property of local intersection multiplicities.

Lemma 19. *Let $Q \in \mathbb{R}[x, y]$ be irreducible and suppose that $\mathcal{Z}_{\mathbb{R}}(Q)$ is infinite. Let $f \in \mathbb{R}[x, y] \setminus (Q)$ satisfy $f|_{\mathcal{Z}_{\mathbb{R}}(Q)} \geq 0$. Then every real affine point $z \in \mathcal{Z}_{\mathbb{R}}(Q) \cap \mathcal{Z}_{\mathbb{R}}(f)$ has even complex local intersection multiplicity: $I_z(Q, f) \in 2\mathbb{N}$. In particular, $I_z(Q, f) \geq 2$.*

Proof. Let $\nu : \tilde{C} \rightarrow \mathcal{Z}_{\mathbb{C}}(Q)$ be the normalization of the complex curve defined by Q . For a real point z , the points in $\nu^{-1}(z)$ are either fixed by complex conjugation or occur in conjugate pairs.

At a conjugation-fixed point, choose a real local parameter t . Since $f \circ \nu$ is nonnegative on the corresponding real branch, its first nonzero Taylor term has even order. At a conjugate pair $\{\xi, \bar{\xi}\}$, the two vanishing orders agree and therefore their sum is even. Hence $I_z(Q, f) = \sum_{\xi \in \nu^{-1}(z)} \text{ord}_{\xi}(f \circ \nu)$ is even. $\square$

With the preceding lemmas in hand, we prove the trichotomy of Theorem 17 by distinguishing whether the core variety is empty, infinite, or finite.

Proof of Theorem 17. Set $C := \mathcal{Z}_{\mathbb{R}}(P)$. Since $M(n)$ is P -pure,

$$\{q \in \mathbb{R}[x, y]_{\leq n} : q(X, Y) = \mathbf{0}\} = P \mathbb{R}[x, y]_{\leq n-3}.$$

Consequently,

$$\text{rank } M(n) = \dim \mathbb{R}[x, y]_{\leq n} - \dim \mathbb{R}[x, y]_{\leq n-3} = \binom{n+2}{2} - \binom{n-1}{2} = 3n.$$

We distinguish three cases. If $\mathcal{CV}(\beta) = \emptyset$, then the Core Variety Theorem, Theorem 5, implies that β has no representing measure. This is alternative (iii).

Suppose next that $\mathcal{CV}(\beta)$ is infinite. By Lemma 18(i), there exists $F \in \mathbb{R}[x, y]_{\leq 2n}$ such that $\mathcal{CV}^{(1)}(\beta) = C \cap \mathcal{Z}_{\mathbb{R}}(F)$. Since $\mathcal{CV}(\beta) \subseteq \mathcal{CV}^{(1)}(\beta)$ is infinite, Bézout's theorem gives $P \mid F$. Hence $\mathcal{CV}^{(1)}(\beta) = C$, and the defining family is unchanged at every subsequent iteration. Therefore $\mathcal{CV}(\beta) = C$. The Core Variety Theorem guarantees a representing measure, so alternative (i) holds.

It remains to suppose that $\mathcal{CV}(\beta)$ is finite and nonempty. By the Core Variety Theorem, β admits a finitely atomic representing measure μ . By Lemma 18(ii), there exists $f \in \mathbb{R}[x, y]_{\leq 2n}$ such that $f \notin (P)$, $f|_C \geq 0$, $L_\beta(f) = 0$, and $\mathcal{CV}(\beta) \subseteq \mathcal{CV}^{(1)}(\beta) = C \cap \mathcal{Z}_\mathbb{R}(f)$.

Let $d := \deg f \leq 2n$, and let P^h and f^h denote the homogenizations of P and f . Since $f \notin (P)$, the projective curves defined by P^h and f^h have no common component. Bézout's theorem and Lemma 19 therefore give

$$2 \operatorname{card}(C \cap \mathcal{Z}_\mathbb{R}(f)) \leq \sum_{z \in \mathcal{Z}_{\mathbb{P}^2_\mathbb{C}}(P^h) \cap \mathcal{Z}_{\mathbb{P}^2_\mathbb{C}}(f^h)} I_z(P^h, f^h) = 3d \leq 6n.$$

Hence

$$\operatorname{card}(C \cap \mathcal{Z}_\mathbb{R}(f)) \leq 3n.$$

Every representing measure is supported on the core variety. Consequently,

$$\operatorname{supp} \mu \subseteq \mathcal{CV}(\beta) \subseteq C \cap \mathcal{Z}_\mathbb{R}(f),$$

and therefore

$$3n = \operatorname{rank} M(n) \leq \operatorname{card}(\operatorname{supp} \mu) \leq \operatorname{card} \mathcal{CV}(\beta) \leq \operatorname{card}(C \cap \mathcal{Z}_\mathbb{R}(f)) \leq 3n.$$

All inequalities are thus equalities. In particular,

$$\operatorname{card} \mathcal{CV}(\beta) = \operatorname{rank} M(n) = 3n.$$

By [13, Theorem 2.20], β has a unique representing measure and its support is precisely $\mathcal{CV}(\beta)$. This is alternative (ii). Finally, if P is affinely equivalent to $y - x^3$, the impossibility of alternative (ii) follows from [14, Theorem 4.1], together with affine invariance of the truncated moment problem. This completes the proof. $\square$

We next relate the trichotomy to the concrete solvability criteria in [15] and give degree-six examples of the finite-core alternative for several affine types. Given a vector $v = (v_1, \dots, v_{2k+1})^t \in \mathbb{R}^{2k+1}$, we denote by

$$H_v := (v_{i+j-1})_{i,j=1}^{k+1}$$

the associated $(k+1) \times (k+1)$ Hankel matrix.

Let $I \subset \mathbb{R}[x, y]$ be the ideal generated by P , denoted by $I = (P)$, and set

$$I_{2n} := I \cap \mathbb{R}[x, y]_{\leq 2n}.$$

Let $\beta \equiv \beta^{(2n)}$ be a P -pure sequence, and let $L_\beta : \mathbb{R}[x, y]_{\leq 2n} \rightarrow \mathbb{R}$ denote its Riesz functional. Then the induced functional

$$\tilde{L}_\beta : \mathbb{R}[x, y]_{\leq 2n} / I_{2n} \rightarrow \mathbb{R}, \quad \text{defined by} \quad \tilde{L}_\beta(p + I_{2n}) := L_\beta(p),$$

is well defined: if $h \in (P) \cap \mathbb{R}[x, y]_{\leq 2n}$, write each monomial multiple of P as Pab with $\deg a \leq n-3$ and $\deg b \leq n$; then P -purity gives $(Pa)(X, Y) = \mathbf{0}$, and hence $L_\beta(Pab) = 0$.

We examine the relevant irreducible normal forms in separate subsections.

Standing Assumption: $M(n)$ is P -pure and positive semidefinite.	(SA)
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We shall use the same perturbation argument in each full-core case below. For $w \in \mathcal{Z}_{\mathbb{R}}(P)$, P -purity implies $\mathbf{v}_n(w)^t \in \text{Ran } M(n)$. Hence Lemma 7 shows that $M(n) - \rho P_n(w)$ is positive semidefinite and remains P -pure for all sufficiently small $\rho > 0$. Any auxiliary matrix required to be positive definite remains so by openness. Once the relevant solvability criterion is applied to the perturbed data, adding $\rho \delta_w$ produces a representing measure for the original sequence whose support contains w .

4.1. Smooth Weierstraß curves: $y^2 - x(x-a)(x-b)$ or $y^2 - x(x^2 + c^2)$, where $0 < a < b$ and $c \neq 0$. When $M(n)$ has a column relation associated with these curves, all moments $\beta_{3,j} = L_{\beta}(x^3 y^j)$ ($0 \leq j \leq 2n-3$) are determined since $M(n)$ is recursively generated. Having excluded these moments, we focus instead on the remaining ones and now define

$$\begin{aligned} u^{(n)} &:= (u_1, \dots, u_{3n})^t \\ &= (1, x, y, \underbrace{x^2, xy, y^2}, \dots, \underbrace{x^2 y^{i-2}, xy^{i-1}, y^i}, \dots, \underbrace{x^2 y^{n-2}, xy^{n-1}, y^n})^t, \\ v^{(n)} &:= (v_1, \dots, v_{3n})^t \\ &= (1, y/x, x, y, \underbrace{x^2, xy, y^2}, \dots, \underbrace{x^2 y^{i-2}, xy^{i-1}, y^i}, \dots, \underbrace{x^2 y^{n-2}, xy^{n-1}})^t, \end{aligned} \quad (4.1)$$

and set

$$M^{(n)} := \left[\tilde{L}_{\beta}(u_i u_j) \right]_{i,j=1}^{3n}, \quad N^{(n)} := \left[\tilde{L}_{\beta}(x v_i v_j) \right]_{i,j=1}^{3n}, \quad (4.2)$$

which are called *localizing matrices* of $M(n)$.

We introduce the C -degree by $\deg_C(x^i y^j) := 2i + 3j$, and extend it to the rational monomial y/x by defining $\deg_C(y/x) := \deg_C(y) - \deg_C(x) = 1$. More generally, for monomials p, q we may set $\deg_C(p/q) := \deg_C(p) - \deg_C(q)$. For a nonzero polynomial $f = \sum_{i,j} a_{i,j} x^i y^j \in \mathbb{R}[x, y]$, we can define $\deg_C f := \max\{\deg_C(x^i y^j) : a_{i,j} \neq 0\}$.

The C -degree extends naturally to expressions involving y/x by working in the ring $\mathbb{R}[x, y, y/x] \subset \mathbb{R}[x^{\pm 1}, y]$. For $i, j, k \in \mathbb{Z}_{\geq 0}$, we have $\deg_C(x^i y^j (y/x)^k) = 2i + 3j + k$. For a nonzero element $g = \sum_{i,j,k} a_{i,j,k} x^i y^j (y/x)^k$, we finally set

$$\deg_C g := \max\{2i + 3j + k : a_{i,j,k} \neq 0\}.$$

If $N^{(n)}$ (see (4.2)) is positive semidefinite and singular, let $w_{\text{lgen}} \in \ker N^{(n)}$ be a vector such that $p_{\text{lgen}} := w_{\text{lgen}}^t v^{(n)}$ is monic and of minimal C -degree among all nonzero polynomials of the form $w^t v^{(n)}$ with $w \in \ker N^{(n)}$. Such a w_{lgen} exists and is unique; we call p_{lgen} the *generating polynomial* of L_{β} .

By [15, Section 4.1], a P -pure sequence β admits a $\mathcal{Z}_{\mathbb{R}}(P)$ -representing measure if and only if $M^{(n)}$ is positive definite and one of the following conditions holds:

- (i) $N^{(n)}$ is positive definite; or
- (ii) $N^{(n)}$ is positive semidefinite and the unique extension of L_{β} to $\mathbb{R}[x, y]_{\leq 2n+2}$, which we also denote by L_{β} , defined by

$$L_{\beta}(y p_{\text{lgen}}) = L_{\beta}(x^{i+1} y^j p_{\text{lgen}}) = 0,$$

for $i \in \{0, 1, 2\}$, $j \in \mathbb{Z}_+$, and $6n - 1 \leq 2i + 3j + \deg_C(p_{\text{lgen}}) \leq 6n + 4$, satisfies $M^{(n+1)} \succeq 0$ and $N^{(n+1)} \succeq 0$.

If (i) holds, then $\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(P)$. Indeed, perturbing β by a sufficiently small multiple of any point $(x, y) \in \mathcal{Z}_{\mathbb{R}}(P)$ preserves the positive semidefiniteness of $M(n)$ as well as the positive definiteness of $M^{(n)}$ and $N^{(n)}$. The perturbed sequence therefore admits a representing measure, and hence the original sequence admits a representing measure whose support contains (x, y) .

If (ii) holds, then the extension of L_β to $\mathbb{R}[x, y]_{\leq 2n+2}$ is unique, and any representing measure has exactly $3n$ atoms by the first part of Theorem 17, since the core variety is finite in this case. The following example shows that this situation can indeed occur.

Example 20. Let $P(x, y) = y^2 - x(x^2 + 3)$, set $R := \{\frac{3}{16}, 1, 3, 12\}$, and define

$$S := \{(0, 0)\} \cup \{(r, \pm\sqrt{r(r^2 + 3)}) : r \in R\}.$$

The accompanying computations were performed in *Mathematica* [17] and are available online.¹ Let $\mu_0 := \sum_{z \in S} \delta_z$, and let $\beta^{(6)}$ be its truncated moment sequence; equivalently, $\beta_{ij} = \sum_{(x,y) \in S} x^i y^j$ for $i + j \leq 6$. In particular, $\beta_{ij} = 0$ whenever j is odd.

The restrictions of $u_1, \dots, u_9$ (see (4.1)) to $\mathcal{Z}_{\mathbb{R}}(P)$ have the form $A(x) + yB(x)$, where $\deg A \leq 4$ and $\deg B \leq 3$. If such a function vanishes on S , the pairs of points with y -coordinates of opposite signs imply that A vanishes on $\{0, \frac{3}{16}, 1, 3, 12\}$ and B vanishes on R . Hence $A = B = 0$, so $M^{(3)} \succ 0$. Since $P(X, Y) = \mathbf{0}$ and the corresponding nine quotient columns are linearly independent, $M(3) \succeq 0$ is P -pure and $\text{rank } M(3) = 9$.

Because $x \geq 0$ on $\mathcal{Z}_{\mathbb{R}}(P)$, the matrix $N^{(3)}$ is positive semidefinite. An exact kernel computation gives $\text{rank } N^{(3)} = 8$ and $p_{\text{lgen}} = xy^2 - \frac{259}{16}y^2 + 51x^2 + 3x + \frac{27}{4}$. Modulo P , this generating polynomial reduces to $p_{\text{lgen}} \equiv \frac{1}{16}(x-1)(x-3)(x-12)(16x-3) \pmod{(P)}$. Thus p_{lgen} vanishes at the eight atoms of $S \setminus \{(0, 0)\}$. At the remaining atom, $x = y = 0$; therefore μ_0 also induces the extension required in (ii), for which $M^{(4)} \succeq 0$ and $N^{(4)} \succeq 0$.

Finally, if σ is any representing measure for $\beta^{(6)}$, then $0 = L_\beta(xp_{\text{lgen}}^2) = \int x p_{\text{lgen}}^2 d\sigma$. The integrand is nonnegative on $\mathcal{Z}_{\mathbb{R}}(P)$, so $\text{supp } \sigma \subseteq \mathcal{Z}_{\mathbb{R}}(P) \cap (\mathcal{Z}_{\mathbb{R}}(x) \cup \mathcal{Z}_{\mathbb{R}}(p_{\text{lgen}})) = S$. Since $\text{rank } M(3) = 9 = \text{card } S$, every representing measure must be supported on all nine points of S , and its weights are uniquely determined. Consequently, μ_0 is the unique 9-atomic representing measure for $\beta^{(6)}$, and $\mathcal{CV}(\beta^{(6)}) = S \subsetneq \mathcal{Z}_{\mathbb{R}}(P)$.

4.2. Neile's parabola: $y^2 - x^3$. For $m \in \{0, 2, 3, \dots, 6n\}$, define γ_m by

$$\gamma_m := \begin{cases} \beta_{0, m/3}, & \text{if } m \equiv 0 \pmod{3}, \\ \beta_{2, \lfloor m/3 \rfloor - 1}, & \text{if } m \equiv 1 \pmod{3}, \\ \beta_{1, \lfloor m/3 \rfloor}, & \text{if } m \equiv 2 \pmod{3}. \end{cases}$$

¹See <https://github.com/ZalarA/TMP-core-variety>.

Recall the standing assumption (SA). By [18, Corollary 4.4], β admits a representing measure supported on $\mathcal{Z}_{\mathbb{R}}(P)$ if and only if $H_{(\gamma_2, \dots, \gamma_{6n-2})}$ is positive definite and $H_{(\gamma_2, \dots, \gamma_{6n})}$ is positive semidefinite.

If $H_{(\gamma_2, \dots, \gamma_{6n})}$ is positive definite, then $\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(P)$. Indeed, perturbing β by a sufficiently small multiple of any point $(x, y) \in \mathcal{Z}_{\mathbb{R}}(P)$ preserves the positive semidefiniteness of $M(n)$ as well as the positive definiteness of $H_{(\gamma_2, \dots, \gamma_{6n})}$. The perturbed sequence therefore admits a representing measure, and hence the original sequence admits a representing measure whose support contains (x, y) .

If $H_{(\gamma_2, \dots, \gamma_{6n})}$ is positive semidefinite and singular, then [3, Theorem 3.9] implies that γ admits a unique representing measure supported on the real and distinct zeros of

$$p(t) := t^{3n} - (t, t^2, \dots, t^{3n-1}) \left(H_{(\gamma_2, \dots, \gamma_{6n-2})} \right)^{-1} (\gamma_{3n+1}, \dots, \gamma_{6n-1})^t.$$

Consequently, $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n) = 3n$. The following example shows that this case can indeed occur.

Example 21. Set $n = 3$ and $P(x, y) = y^2 - x^3$. Define $\beta^{(6)}$ by $\beta_{00} = 9$ and $\beta_{ij} := \sum_{k=1}^8 k^{-(2i+3j)}$ for $1 \leq i+j \leq 6$. Thus $\gamma_0 = 9$ and $\gamma_m = \sum_{k=1}^8 k^{-m}$ for $m = 2, \dots, 18$.

The sequence $\beta^{(6)}$ is represented by the nine points obtained from $t = 0, 1, 1/2, \dots, 1/8$ under the parametrization $t \mapsto (t^2, t^3)$; hence $M(3) \succeq 0$. Moreover, $M(3)$ has rank 9. Indeed, if $q(t) = c_0 + \sum_{m=2}^9 c_m t^m$ vanishes at $0, 1, 1/2, \dots, 1/8$, then $c_0 = 0$ and $q(t) = t^2 r(t)$, where $\deg r \leq 7$. Since r has the eight distinct zeros $1, 1/2, \dots, 1/8$, it follows that $q = 0$. Therefore the only degree-3 column relation is $P(X, Y) = \mathbf{0}$, so $\beta^{(6)}$ is P -pure.

The matrix $H_{(\gamma_2, \dots, \gamma_{16})}$ is positive definite because it equals AA^t , where $A = (k^{-r})_{r,k=1}^8$ is a nonsingular Vandermonde-type matrix. Similarly, $H_{(\gamma_2, \dots, \gamma_{18})} = BB^t$, where $B = (k^{-r})_{\substack{1 \leq r \leq 9 \\ 1 \leq k \leq 8}}$ has rank 8. Hence $H_{(\gamma_2, \dots, \gamma_{18})}$ is positive semidefinite and singular.

The corresponding generating polynomial is $p(t) = t \prod_{k=1}^8 (t - \frac{1}{k})$, whose nine zeros are real and distinct. It follows from the preceding criterion that $\nu = \delta_0 + \sum_{k=1}^8 \delta_{1/k}$ is the unique representing measure for γ . Consequently, its pushforward under $t \mapsto (t^2, t^3)$, namely $\mu = \delta_{(0,0)} + \sum_{k=1}^8 \delta_{(k^{-2}, k^{-3})}$, is the unique 9-atomic representing measure for $\beta^{(6)}$.

4.3. Nodal cubic: $y^2 - x(x-1)^2$. Let

$$V_{\leq 6n} := \{ s \in \mathbb{R}[t]_{\leq 6n} : s(1) = s(-1) \},$$

and define a linear map

$$\Phi : \mathbb{R}[x, y]_{\leq 2n} / I_{2n} \rightarrow V_{\leq 6n} \quad \text{by} \quad \Phi(p + I_{2n}) := p(t^2, t^3 - t).$$

A dimension count shows that Φ is a vector space isomorphism. Define a linear functional $L : V_{\leq 6n} \rightarrow \mathbb{R}$ by

$$L(s) := \tilde{L}_{\beta}(\Phi^{-1}(s)).$$

Then β admits a $\mathcal{Z}_{\mathbb{R}}(P)$ -representing measure if and only if L admits a representing measure on $\mathbb{R}$. Indeed, suppose that $\mu = \sum_{\ell} \alpha_{\ell} \delta_{(x_{\ell}, y_{\ell})}$ represents β . For each ℓ , choose

$t_\ell \in \mathbb{R}$ such that $(x_\ell, y_\ell) = (t_\ell^2, t_\ell^3 - t_\ell)$. Explicitly, set $t_\ell = \frac{y_\ell}{x_\ell - 1}$ if $(x_\ell, y_\ell) \neq (1, 0)$. If $(x_\ell, y_\ell) = (1, 0)$, one may choose either $t_\ell = 1$ or $t_\ell = -1$, since every $s \in V_{\leq 6n}$ satisfies $s(1) = s(-1)$. Then $\sum_\ell \alpha_\ell \delta_{t_\ell}$ is a representing measure for L . Conversely, if $\sum_\ell \alpha_\ell \delta_{t_\ell}$ represents L , then $\sum_\ell \alpha_\ell \delta_{(t_\ell^2, t_\ell^3 - t_\ell)}$ represents β .

Let

$$\begin{aligned} u &:= (u_1, \dots, u_{3n}) = (1, t^2 - 1, t^3 - t, \dots, t^{3n} - t^{3n-2}), \\ v &:= (v_1, \dots, v_{3n}) = (t, t^2 - 1, t^3 - t, \dots, t^{3n} - t^{3n-2}), \end{aligned}$$

and define

$$M := [L(u_i u_j)]_{i,j=1}^{3n}, \quad N := [L(v_i v_j)]_{i,j=1}^{3n} = \begin{bmatrix} N_1 & \mathbf{n}_1 \\ \mathbf{n}_1^t & L(v_{3n}^2) \end{bmatrix}. \quad (4.3)$$

Recall the standing assumption (SA). For P -pure sequences β , it follows from [15, Section 4.3] that β admits a $\mathcal{Z}_{\mathbb{R}}(P)$ -representing measure if and only if M and N_1 are positive definite and N is positive semidefinite.

If N is positive definite, then $\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(P)$, since perturbing β by a sufficiently small multiple of any point $(x, y) \in \mathcal{Z}_{\mathbb{R}}(P)$ preserves the positive definiteness of both M and N .

If N is positive semidefinite and singular, then by the proof of [15, Theorem 4.13, Case 2], the functional L admits a unique representing measure supported on the zeros of

$$p(t) := t^{3n} - t^{3n-2} - (t, t^2 - 1, t^3 - t, \dots, t^{3n-1} - t^{3n-3}) N_1^{-1} (L(v_1 v_{3n}), \dots, L(v_{3n-1} v_{3n}))^t.$$

Consequently, $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n) = 3n$. The following example shows that this case can indeed occur.

Example 22. Set $n = 3$. For $f \in V_{\leq 18}$, let $[t^{18}]f$ denote the coefficient of t^{18} in f , and define $L(f) := f(0) + \sum_{j=1}^5 (f(j) + f(-j)) - \frac{131681894400}{2431} [t^{18}]f$. Let $\beta^{(6)}$ be the truncated moment sequence associated with L through the isomorphism Φ .

The correction affects only the common lower-right Gram entry, since $u_9^2 = v_9^2 = (t^9 - t^7)^2$ is the only product occurring in M or N with a nonzero t^{18} -coefficient. In particular, $L((t^9 - t^7)^2) = \frac{17387981059025664}{2431}$. An exact rational computation in *Mathematica* verifies that $M \succ 0$, $N_1 \succ 0$, $N \succeq 0$, and $\text{rank } N = 8$.

The corresponding generating polynomial is

$$p(t) = t \left(t^8 - \frac{870}{17} t^6 + \frac{14217}{17} t^4 - \frac{1068260}{221} t^2 + \frac{17460288}{2431} \right).$$

It has nine distinct real zeros $0, \pm a_1, \dots, \pm a_4$, where

$$(a_1, a_2, a_3, a_4) \approx (1.496830, 2.844771, 3.980890, 4.999576).$$

The mass at 0 and the equal masses at each pair $\{\pm a_j\}$ are approximately

$$(1.523736, 1.443895, 1.239072, 1.052979, 1.002187),$$

respectively. Enumerate the zeros of p as $t_1, \dots, t_9$, and denote their corresponding positive weights by $w_1, \dots, w_9$. Then $\nu = \sum_{i=1}^9 w_i \delta_{t_i}$ is the unique representing measure for L . Consequently, $\mu = \sum_{i=1}^9 w_i \delta_{(t_i^2, t_i^3 - t_i)}$ is the unique 9-atomic representing measure for $\beta^{(6)}$.

4.4. **Isolated point:** $y^2 - x^2(x - 1)$. Let

$$V_{\leq 6n} := \{s \in \mathbb{R}[t]_{\leq 6n} : s(\mathbf{i}) = s(-\mathbf{i})\},$$

and define a linear map

$$\Phi : \mathbb{R}[x, y]_{\leq 2n} / I_{2n} \rightarrow V_{\leq 6n} \quad \text{by} \quad \Phi(p + I_{2n}) := p(t^2 + 1, t^3 + t).$$

As before, Φ is a vector space isomorphism. Define a linear functional $L : V_{\leq 6n} \rightarrow \mathbb{R}$ by

$$L(s) := \tilde{L}_\beta(\Phi^{-1}(s)).$$

Then β admits a $(\mathcal{Z}_\mathbb{R}(P) \setminus \{(0, 0)\})$ -representing measure if and only if L admits a representing measure on $\mathbb{R}$.

Let

$$u := (u_1, \dots, u_{3n}) = (1, t^2 + 1, t^3 + t, \dots, t^{3n} + t^{3n-2}),$$

$$v := (v_1, \dots, v_{3n}) = (t, t^2 + 1, t^3 + t, \dots, t^{3n} + t^{3n-2}),$$

and define

$$M := [L(u_i u_j)]_{i,j=1}^{3n} = \begin{bmatrix} L(1) & \mathbf{m}_1^t \\ \mathbf{m}_1 & M_1 \end{bmatrix}, \quad (4.4)$$

$$N := [L(v_i v_j)]_{i,j=1}^{3n} = \begin{bmatrix} N_1 & \mathbf{n}_1 \\ \mathbf{n}_1^t & L(v_{3n}^2) \end{bmatrix} = \begin{bmatrix} L(t^2) & \mathbf{n}_2^t \\ \mathbf{n}_2 & N_2 \end{bmatrix}. \quad (4.5)$$

With the block decompositions in (4.4)–(4.5), the Schur complements appearing below are

$$M/M_1 = L(1) - \mathbf{m}_1^t M_1^{-1} \mathbf{m}_1, \quad N/N_2 = L(t^2) - \mathbf{n}_2^t N_2^\dagger \mathbf{n}_2.$$

Recall the standing assumption (SA). For P -pure sequences β , [15, Theorem 4.18] implies that β admits a $\mathcal{Z}_\mathbb{R}(P)$ -representing measure if and only if M is positive definite, N_2 is positive semidefinite, $n_2 \in \text{Ran } N_2$, and one of the following conditions holds:

- (i) $N \succ 0$;
- (ii) $M/M_1 > -N/N_2 \geq 0$;
- (iii) $M/M_1 = -N/N_2$ and

$$\tilde{L} := L - \frac{M/M_1}{2}(L_{(-\mathbf{i})} + L_{(\mathbf{i})}) \quad (4.6)$$

is a moment functional, where $\mathbf{i}$ stands for the imaginary unit and $L_{(\mathbf{i})}, L_{(-\mathbf{i})} : V_{\leq 6n} \rightarrow \mathbb{C}$ are linear functionals defined by $L_{(\mathbf{i})}(f) := f(\mathbf{i})$, $L_{(-\mathbf{i})}(f) := f(-\mathbf{i})$, $f \in V_{\leq 6n}$.

If (i) holds, then $\mathcal{CV}(\beta) = \mathcal{Z}_\mathbb{R}(P)$, since perturbing β by a sufficiently small multiple of any point $(x, y) \in \mathcal{Z}_\mathbb{R}(P)$ preserves the positive definiteness of both M and N .

If (ii) holds, replacing $L(1)$ by $L(1) - \frac{1}{2}(M/M_1 - N/N_2)$ yields a functional satisfying (i). Moreover, a representing measure then contains the atom $(0, 0)$.

If (iii) holds, then $(0, 0)$ is forced as an atom of mass M/M_1 . In this case, if $\tilde{L}$ admits a representing measure, it is unique and supported on $3n - 1$ atoms. Consequently,

$\text{card } \mathcal{CV}(\beta) = \text{rank } M(n) = 3n$. The following example shows that this case can indeed occur.

Example 23. Consider the isolated-point cubic $P(x, y) = y^2 - x^2(x - 1)$. Define a real linear functional $L : \mathbb{R}[t] \rightarrow \mathbb{R}$ by

$$L(f) := \sum_{j=1}^4 (f(j) + f(-j)) + f(\mathbf{i}) + f(-\mathbf{i}), \quad f \in \mathbb{R}[t],$$

where $\mathbf{i}$ denotes the imaginary unit, and let $\beta^{(6)}$ be the truncated moment sequence associated with $L|_{V_{\leq 18}}$ through the isomorphism Φ defined above. This compact prescription gives all the moment values required to construct M and N .

An exact-arithmetic computation in *Mathematica* verifies that $M \succ 0$, $N_2 \succ 0$, and $M/M_1 = -N/N_2 = [2]$. Since $M/M_1 = 2$, the functional in (4.6) satisfies

$$\tilde{L}(f) = L(f) - f(\mathbf{i}) - f(-\mathbf{i}) = \sum_{j=1}^4 (f(j) + f(-j)).$$

Hence $\tilde{L}$ has the 8-atomic representing measure $\nu = \sum_{j=1}^4 (\delta_j + \delta_{-j})$. Thus condition (iii) above holds, and the preceding analysis shows that ν is the unique representing measure for $\tilde{L}$, while $(0, 0)$ is a forced atom of mass 2 for $\beta^{(6)}$.

Under the parametrization $t \mapsto (t^2 + 1, t^3 + t)$, the parameters $\pm j$ yield the eight points $(j^2 + 1, \pm(j^3 + j))$, whereas $\pm \mathbf{i}$ both correspond to the isolated point $(0, 0)$. Consequently,

$$\mu = 2\delta_{(0,0)} + \sum_{j=1}^4 (\delta_{(j^2+1, j^3+j)} + \delta_{(j^2+1, -j^3-j)})$$

is the unique 9-atomic representing measure for $\beta^{(6)}$.

4.5. Rational cubic type 1: $P(x, y) = xy^2 + ay - bx - c$, where $a, b, c \in \mathbb{R}$ such that $a^2b \neq c^2$. Define the vector space $V_{\leq 6n}$ by

$$V_{\leq 6n} := \begin{cases} \left\{ \frac{q(t)}{(t^2 - b)^{2n}} : q \in \mathbb{R}[t]_{\leq 6n}, \begin{matrix} (c + ia\sqrt{-b})^{2n} q(\mathbf{i}\sqrt{-b}) = \\ (c - ia\sqrt{-b})^{2n} q(-\mathbf{i}\sqrt{-b}) \end{matrix} \right\}, & b < 0, \\ \left\{ \frac{q(t)}{t^{4n}} : q \in \mathbb{R}[t]_{\leq 6n}, q'(0) = -\frac{2na}{c} q(0) \right\}, & b = 0, \\ \left\{ \frac{q(t)}{(t^2 - b)^{2n}} : q \in \mathbb{R}[t]_{\leq 6n}, \begin{matrix} (c + a\sqrt{b})^{2n} q(\sqrt{b}) = \\ (c - a\sqrt{b})^{2n} q(-\sqrt{b}) \end{matrix} \right\}, & b > 0. \end{cases} \quad (4.7)$$

Define a linear map

$$\Phi : \mathbb{R}[x, y]_{\leq 2n} / I_{2n} \rightarrow V_{\leq 6n} \quad \text{by} \quad \Phi(p + I_{2n}) := p\left(\frac{-at + c}{t^2 - b}, t\right).$$

Then Φ is a vector space isomorphism. Defining

$$L(s) := \tilde{L}_\beta(\Phi^{-1}(s)),$$

we have that β admits a $\mathcal{Z}_{\mathbb{R}}(P)$ -representing measure if and only if L admits a representing measure on $\mathbb{R}$.

Let

$$h(t) := \frac{-at + c}{t^2 - b}.$$

With the moments $\beta_{i+1,j+2} = L_{\beta}(x^{i+1}y^{j+2})$ ($0 \leq i + j \leq 2n - 3$) already determined recursively, we turn our attention to the rest and define

$$\begin{aligned} u &:= (u_1, \dots, u_{3n}) = (h^n, h^{n-1}, h^{n-1}t, \dots, h, ht, 1, t, \dots, t^n), \\ v &:= (v_1, \dots, v_{3n}) = (h^nt, h^{n-1}, h^{n-1}t, \dots, h, ht, 1, t, \dots, t^n). \end{aligned} \quad (4.8)$$

Set

$$M := [L(u_i u_j)]_{i,j=1}^{3n}, \quad N := [L(v_i v_j)]_{i,j=1}^{3n} = \begin{bmatrix} N_1 & \mathbf{n}_1 \\ \mathbf{n}_1^t & L(v_{3n}^2) \end{bmatrix}. \quad (4.9)$$

Recall the standing assumption (SA). For P -pure sequences β , it follows from [15, Section 5.1] that β admits a $\mathcal{Z}_{\mathbb{R}}(P)$ -representing measure if and only if M and N_1 are positive definite, N is positive semidefinite, and, in the case $b > 0$ with N singular, the functional

$$L_{\mathbb{R}[t]_{\leq 6n-2}} : \mathbb{R}[t]_{\leq 6n-2} \rightarrow \mathbb{R}, \quad L_{\mathbb{R}[t]_{\leq 6n-2}}(f) := L\left(\frac{f}{(t^2 - b)^{2n-2}}\right),$$

does not place mass at $\pm\sqrt{b}$ in its unique representing measure.

If $N \succ 0$, then

$$\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(P),$$

since perturbing β by a sufficiently small multiple of any point $(x, y) \in \mathcal{Z}_{\mathbb{R}}(P)$ preserves the positive definiteness of both M and N .

If $N \succeq 0$ is singular, then by the proofs of [15, Theorems 5.6 and 5.7], the functional L admits a unique representing measure supported on the zeros of

$$p(t) := (t^2 - b)^n \left(t^n - (h^nt, h^{n-1}, h^{n-1}t, \dots, h, ht, 1, \dots, t^{n-1}) N_1^{-1} \mathbf{n}_1 \right). \quad (4.10)$$

Consequently, $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n) = 3n$. The following example shows that this case can indeed occur.

Example 24. Let $P(x, y) = xy^2 - x + y$. Here $n = 3$, $a = b = 1$, and $c = 0$. Set $d(t) := t^2 - 1$ and $h(t) := -t/d(t)$. Every $f \in V_{\leq 18}$ has a unique representation $f = q/d^6$, where $q \in \mathbb{R}[t]_{\leq 18}$ and $q(1) = q(-1)$.

Let $\tau := 23938/24013$, $S := \{\pm 2, \pm 3, \pm 4, \pm 5, \tau\}$, and define $L_0(f) := \sum_{s \in S} f(s)$. For $f = q/d^6$, let $\lambda_{18}(f) := [t^{18}]q(t)$, where $[t^{18}]q$ denotes the coefficient of t^{18} in q , and define $L(f) := L_0(f) - \frac{1881600}{7951} \lambda_{18}(f)$.

Let $u = (u_1, \dots, u_9)$ and $v = (v_1, \dots, v_9)$ be as in (4.8). Among the products $u_i u_j$ and $v_i v_j$, only $u_9^2 = v_9^2 = t^6$ has a nonzero t^{18} -coefficient after multiplication by d^6 . Thus L agrees with L_0 on all the other entries of M and N , while $L(t^6) = L_0(t^6) - 1881600/7951$. This correction is chosen so that the Schur complement of N_1 in N vanishes.

An exact rational computation in *Mathematica* verifies that $M \succ 0$, $N_1 \succ 0$, $N \succeq 0$, and $\text{rank } N = 8$. Exact real-root isolation further shows that the degree-9 polynomial $p(t)$ defined in (4.10) has nine distinct real zeros, none equal to ± 1 . In increasing order, they are approximately

$$\begin{aligned} t_1 &\approx -4.828162, & t_2 &\approx -3.216861, & t_3 &\approx -2.003877, \\ t_4 &\approx -1.003091, & t_5 &\approx 0.006725, & t_6 &\approx 0.996877, \\ t_7 &\approx 2.003871, & t_8 &\approx 3.216821, & t_9 &\approx 4.828150. \end{aligned}$$

Hence the exceptional-pole condition in the preceding singular-case criterion is satisfied. It follows that every representing measure for L is supported on these zeros. Since $\text{rank } M = 9$, the corresponding weights are uniquely determined and strictly positive. Therefore $\nu = \sum_{i=1}^9 w_i \delta_{t_i}$, where $w_i > 0$, is the unique representing measure for L .

Finally, $t \mapsto (h(t), t)$ is a bijection from $\mathbb{R} \setminus \{\pm 1\}$ onto $\mathcal{Z}_{\mathbb{R}}(P)$. Consequently, $\mu = \sum_{i=1}^9 w_i \delta_{(h(t_i), t_i)}$ is the unique 9-atomic representing measure for the associated bivariate truncated moment sequence $\beta^{(6)}$.

4.6. Rational cubic type 2: $xy - x^3 - q_2x^2 - q_1x - q_0$, $q_0 \neq 0$.

Let

$$\gamma = (\gamma_{-2n}(\gamma_{4n}), \gamma_{-2n+1}, \dots, \gamma_{4n})$$

be defined as in [16, Section 3], namely,

$$\begin{aligned} \gamma_i &:= \beta_{i,0}, & i &= 0, \dots, 2n, \\ \gamma_{-k} &:= \frac{1}{q_0^k} \left(\beta_{0,k} - \sum_{t=-k+1}^{2k} q_{k,t} \gamma_t \right), & k &= 1, \dots, 2n-1, \\ \gamma_{2n+k} &:= \beta_{2n-k,k} - \sum_{t=-k}^{2k-1} q_{k,t} \gamma_{t+2n-k}, & k &= 1, \dots, 2n-1, \\ \gamma_{-2n} &:= \frac{1}{q_0^{2n}} \left(\beta_{0,2n} - \sum_{t=-2n+1}^{4n-1} q_{2n,t} \gamma_t - \gamma_{4n} \right) =: D - q_0^{-2n} \gamma_{4n}. \end{aligned}$$

Here the coefficients $q_{j,t}$ are given by

$$q_{j,t} := \sum_{\substack{k_0+k_1+k_2+k_3=j, \\ k_r \in \mathbb{Z}_+, \sum_{s=0}^3 (2-s)k_s=t}} \frac{j!}{k_0! k_1! k_2! k_3!} q_2^{k_1} q_1^{k_2} q_0^{k_3}.$$

Define the Hankel matrix

$$H_{\gamma(t)} = \begin{bmatrix} D - q_0^{-2n} t & \mathbf{b}^t & \gamma_n \\ \mathbf{b} & H_{\tilde{\gamma}} & \mathbf{c} \\ \gamma_n & \mathbf{c}^t & t \end{bmatrix}, \quad (4.11)$$

where

$$\mathbf{b}^t := (\gamma_{-2n+1}, \dots, \gamma_{-n}, \dots, \gamma_{n-1}),$$

$$\begin{aligned}\mathbf{c}^t &:= (\gamma_{n+1}, \dots, \gamma_{2n}, \dots, \gamma_{4n-1}), \\ \tilde{\gamma} &:= (\gamma_{-2n+2}, \dots, \gamma_{4n-2}).\end{aligned}$$

Recall the standing assumption (SA). By [16, Theorem 2.2], β admits a representing measure supported on $\mathcal{Z}_{\mathbb{R}}(P)$ if and only if there exists $t \in \mathbb{R}$ such that $H_{\gamma(t)} \succeq 0$ and either

- (i) $H_{\gamma(t)} \succ 0$, or
- (ii) $\text{rank } H_{\gamma(t)} = \text{rank} \begin{bmatrix} D - q_0^{-2n}t & \mathbf{b}^t \\ \mathbf{b} & H_{\tilde{\gamma}} \end{bmatrix} = \text{rank} \begin{bmatrix} H_{\tilde{\gamma}} & \mathbf{c} \\ \mathbf{c}^t & t \end{bmatrix}$.

Set

$$\begin{aligned}t_{\min} &:= \mathbf{c}^t H_{\tilde{\gamma}}^{-1} \mathbf{c}, \\ t_{\max} &:= q_0^{2n} (D - \mathbf{b}^t H_{\tilde{\gamma}}^{-1} \mathbf{b}), \\ E &:= \left(\mathbf{b}^t H_{\tilde{\gamma}}^{-1} \mathbf{c} \right) \left(\mathbf{b}^t H_{\tilde{\gamma}}^{-1} \mathbf{c} \right)^t - 2\mathbf{b}^t H_{\tilde{\gamma}}^{-1} \mathbf{c} \gamma_n + \gamma_n^2.\end{aligned}$$

By [16, Theorem 3.1], β admits a $\mathcal{Z}_{\mathbb{R}}(P)$ -representing measure if and only if $t_{\min} < t_{\max}$ and $(t_{\max} - t_{\min})^2 \geq 4q_0^{2n}E$, in which case there exists a $3n$ -atomic representing measure.

If $(t_{\max} - t_{\min})^2 > 4q_0^{2n}E$, then $\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(P)$, since perturbing β by a sufficiently small multiple of any point $(x, y) \in \mathcal{Z}_{\mathbb{R}}(P)$ preserves the solvability of the conditions above.

If $(t_{\max} - t_{\min})^2 = 4q_0^{2n}E$, then there exists a unique parameter t for which $H_{\gamma(t)} \succeq 0$. Consequently, β has a unique representing measure, whose x -coordinates are supported on the zeros of the generating polynomial of $H_{\gamma(t)}$. The following example shows that this case can indeed occur.

Example 25. Let $P(x, y) = xy - x^3 - 1$. For $f \in \mathbb{R}[z, z^{-1}]$, define

$$L_0(f) := \sum_{j=1}^4 (f(j) + f(-j)) + 4f(5),$$

and set $D := \frac{35873030698983031183702201}{35478552096000000}$.

We prescribe the data entering $H_{\gamma(\tau)}$ by $\gamma_m := L_0(z^m)$ for $-5 \leq m \leq 11$, together with $\gamma_{-6} := D - \tau$ and $\gamma_{12} := \tau$. Let $\beta^{(6)}$ be the corresponding bivariate truncated moment sequence.

Set $\tau_* := \frac{1537767051345640}{1520857}$ and $\Delta := \frac{36578304}{1520857}$. An exact rational computation gives $t_{\min} = \tau_* - \Delta$, $t_{\max} = \tau_* + \Delta$, and $E = \Delta^2$. In particular, $t_{\min} < t_{\max}$ and $(t_{\max} - t_{\min})^2 = 4E$, so τ_* is the unique parameter for which $H_{\gamma(\tau_*)} \succeq 0$.

At $\tau = \tau_*$, one has $\text{rank } H_{\gamma(\tau_*)} = 9$. An exact computation also verifies that the associated moment matrix $M(3)$ is positive semidefinite, P -pure, and has rank 9. The generating polynomial of $H_{\gamma(\tau_*)}$ has nine distinct nonzero real zeros $x_1, \dots, x_9$. Numerically, its four negative zeros are approximately $-3.98861, -2.85420, -1.75817, -0.980010$, and its five positive zeros are approximately $0.00115558, 1.01588, 2.30066, 3.77819, 4.99602$.

Let $w_1, \dots, w_9 > 0$ be the corresponding uniquely determined weights. Then $\nu = \sum_{i=1}^9 w_i \delta_{x_i}$ is the unique representing measure for γ . Since $P(x, y) = 0$ is parametrized by

$x \mapsto (x, x^2 + x^{-1})$, the measure $\mu = \sum_{i=1}^9 w_i \delta_{(x_i, x_i^2 + x_i^{-1})}$ is the unique 9-atomic representing measure for $\beta^{(6)}$.

4.7. Cubical parabola: $y - x^3$. This case follows from [14, Theorem 4.1].

4.8. Non-Weierstraß types 1–5: The rational transformations used in [15, Theorem 5.8] yield solvability criteria for these normal forms.

5. THE CORE VARIETY $\mathcal{CV}(\beta)$ FOR MOMENT MATRICES $M(n)$ WITH A REDUCIBLE CUBIC RELATION HAVING TWO IRREDUCIBLE FACTORS

In this section, we classify the core varieties arising when the cubic relation factors as a line times an irreducible conic. If a representing measure μ is supported on the union $C_1 \cup C_2$, it can be written as $\mu = \mu_1 + \mu_2$, with $\text{supp } \mu_i \subseteq C_i$; at a common point of the components, its mass may be assigned to either summand. Accordingly,

$$M(n)[\mu] = M(n)[\mu_1] + M(n)[\mu_2].$$

The ranks contributed by the line and conic components are at most $n + 1$ and $2n + 1$, respectively, by Lemma 8. These bounds, together with the first-core certificate, will control all possible recombinations of the component measures.

We begin with the case in which the cubic consists of a line and a conic intersecting at a pair of non-real points.

Theorem 26. *Let $n \geq 3$, let $P = P_1 P_2 \in \mathbb{R}[x, y]$, where $\deg P_1 = 1$, $\deg P_2 = 2$, and P_2 is irreducible with infinite real zero set. Assume that $\text{card}(\mathcal{Z}_{\mathbb{C}}(P_1) \cap \mathcal{Z}_{\mathbb{C}}(P_2)) = 2$ and $\mathcal{Z}_{\mathbb{R}}(P_1) \cap \mathcal{Z}_{\mathbb{R}}(P_2) = \emptyset$. Let $M(n)(\beta)$ be a positive semidefinite moment matrix that is P -pure. Then exactly one of the following alternatives holds:*

- (i) $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = \mathcal{Z}_{\mathbb{R}}(P)$, and a representing measure exists;
- (ii) $\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(P_2) \cup S$, where $S \subseteq \mathcal{Z}_{\mathbb{R}}(P_1)$ and $\text{card } S = n - 1$; a representing measure exists;
- (iii) $\mathcal{CV}(\beta) = \emptyset$, and no representing measure exists.

The next lemma shows that, once an entire component and the maximal possible number of isolated points survive the first core iteration, no subsequent iteration can remove any of them.

Lemma 27. *Let $n \geq 2$ and $\mathcal{CV}(\beta) \neq \emptyset$, put $r := \text{rank } M(n)(\beta)$, and suppose that*

$$\mathcal{CV}^{(1)}(\beta) = D \cup S,$$

where $S \subseteq \mathbb{R}^2 \setminus D$ is finite and D is the real zero set of either a linear polynomial, an irreducible quadratic polynomial with infinite real zero set, or a product of two distinct real linear polynomials. If

$$\rho_n(D) + \text{card } S \leq r,$$

then equality holds and

$$\mathcal{CV}(\beta) = D \cup S.$$

Proof. By the Core Variety Theorem, there is a finitely atomic representing measure μ with $\text{supp } \mu \subseteq \mathcal{CV}(\beta) \subseteq D \cup S$. Write $\mu = \mu_D + \mu_S$, where $\mu_D := \mu|_D$ and $\mu_S := \mu|_S$. Then

$$r \leq \text{rank } M(n)[\mu_D] + \text{rank } M(n)[\mu_S] \leq \rho_n(D) + \text{card}(\text{supp } \mu_S) \leq \rho_n(D) + \text{card } S \leq r.$$

Consequently, all inequalities are equalities. In particular, $\text{supp } \mu_S = S$. We also have $\text{rank } M(n)[\mu_D] = \rho_n(D)$.

Let Q be a polynomial of degree at most two with $D = \mathcal{Z}_{\mathbb{R}}(Q)$. For each of the three types of D listed in the statement, $I(D) = (Q)$. Since $M(n)[\mu_D]$ is supported on D , its kernel contains $Q\mathcal{P}_{n-\deg Q}$. The quotient by this subspace has dimension $\rho_n(D)$; hence the equality $\text{rank } M(n)[\mu_D] = \rho_n(D)$ forces equality of the two kernels. Thus $M(n)[\mu_D]$ is Q -pure. Consequently, Theorem 10 implies that the core variety of the moment sequence generated by μ_D is D . Thus, for every $z \in D$, there is a representing measure ν_z for the moments of μ_D whose support contains z . Since $\nu_z + \mu_S$ represents β , we obtain $D \subseteq \mathcal{CV}(\beta)$. Moreover, $\text{supp } \mu_S = S$ gives $S \subseteq \mathcal{CV}(\beta)$. Therefore $\mathcal{CV}(\beta) = D \cup S$. $\square$

With the preceding rank estimate available, we prove Theorem 26 by first separating the empty-core case and then, when a representing measure exists, analyzing the definiteness of the localizing matrices associated with the two irreducible components.

Proof of Theorem 26. Set $C_i := \mathcal{Z}_{\mathbb{R}}(P_i)$, $C := C_1 \cup C_2$, and $I := (P)$. Since $M(n)(\beta)$ is P -pure, the induced functional $L_C : \mathbb{R}[C]_{\leq 2n} \rightarrow \mathbb{R}$, defined by $L_C(p + I) := L_\beta(p)$, is well defined. Indeed, every monomial of degree at most $2n - 3$ can be written as a product ab with $\deg a \leq n - 3$ and $\deg b \leq n$; P -purity gives $(Pa)(X, Y) = \mathbf{0}$, and hence $L_\beta(Pab) = 0$.

Choose polynomial representatives $f_1, \dots, f_{3n} \in \mathbb{R}[x, y]_{\leq n}$, $g_1, \dots, g_{2n-1} \in \mathbb{R}[x, y]_{\leq n-1}$, and $h_1, \dots, h_n \in \mathbb{R}[x, y]_{\leq n-1}$ such that $\{f_i + I\}_{i=1}^{3n}$, $\{g_i + (P_2)\}_{i=1}^{2n-1}$, and $\{h_i + (P_1)\}_{i=1}^n$ are bases of $\mathbb{R}[C]_{\leq n}$, $\mathbb{R}[C_2]_{\leq n-1}$, and $\mathbb{R}[C_1]_{\leq n-1}$, respectively. Define

$$M^{(n)} := [L_\beta(f_i f_j)]_{i,j=1}^{3n}, \quad M_1^{(n)} := [L_\beta(\chi_1 P_1 g_i g_j)]_{i,j=1}^{2n-1}, \quad M_2^{(n)} := [L_\beta(\chi_2 P_2 h_i h_j)]_{i,j=1}^n,$$

where

$$\chi_1 := \begin{cases} 1, & P_1 > 0 \text{ on } C_2, \\ -1, & P_1 < 0 \text{ on } C_2, \\ 0, & P_1 \text{ takes both positive and negative values on } C_2, \end{cases}$$

$$\chi_2 := \begin{cases} 1, & P_2 > 0 \text{ on } C_1, \\ -1, & P_2 < 0 \text{ on } C_1. \end{cases}$$

These definitions are valid because $C_1 \cap C_2 = \emptyset$; in particular, P_2 has a constant nonzero sign on the connected line C_1 .

By P -purity,

$$\text{rank } M(n)(\beta) = \dim \mathbb{R}[C]_{\leq n} = \dim \mathbb{R}[x, y]_{\leq n} - \dim \mathbb{R}[x, y]_{\leq n-3} = 3n.$$

Moreover, $M^{(n)}$ represents the bilinear form induced by L_β on $\mathbb{R}[C]_{\leq n}$, and hence $M^{(n)} \succ 0$.

If $\mathcal{CV}(\beta) = \emptyset$, the Core Variety Theorem implies that no representing measure exists. Suppose therefore that $\mathcal{CV}(\beta) \neq \emptyset$. By the Core Variety Theorem, β has a C -representing measure μ . Consequently, $M_2^{(n)} \succeq 0$ and, when $\chi_1 \neq 0$, $M_1^{(n)} \succeq 0$.

We first show that $M_1^{(n)}$ cannot be singular when $\chi_1 \neq 0$. Otherwise, there exists $u \in \mathbb{R}[x, y]_{\leq n-1}$, nonzero modulo (P_2) , such that $L_\beta(\chi_1 P_1 u^2) = 0$. The polynomial $\chi_1 P_1 u^2$ is nonnegative on C , vanishes identically on C_1 , and is strictly positive on $C_2 \setminus \mathcal{Z}_\mathbb{R}(u)$. Therefore $\text{supp } \mu \subseteq C_1 \cup (C_2 \cap \mathcal{Z}_\mathbb{R}(u))$. Since $u \notin (P_2)$, Bézout's theorem gives

$$\text{card}(C_2 \cap \mathcal{Z}_\mathbb{R}(u)) \leq 2(n-1).$$

Lemma 8 would then imply $3n = \text{rank } M(n) \leq (n+1) + (2n-2) = 3n-1$, a contradiction. Thus $M_1^{(n)} \succ 0$ whenever $\chi_1 \neq 0$.

Suppose first that $M_2^{(n)} \succ 0$. By [15, Theorem 2.4], every $q \in \mathbb{R}[C]_{\leq 2n}$ that is nonnegative on C admits a representation

$$q = \sum_r a_r^2 + \chi_1 P_1 \sum_s b_s^2 + \chi_2 P_2 \sum_t c_t^2,$$

where $a_r \in \mathbb{R}[C]_{\leq n}$, $b_s \in \mathbb{R}[C_2]_{\leq n-1}$, and $c_t \in \mathbb{R}[C_1]_{\leq n-1}$; the middle sum is absent when $\chi_1 = 0$. Since $M^{(n)}$, $M_2^{(n)}$, and, when present, $M_1^{(n)}$ are positive definite, every nonzero $q \geq 0$ on C satisfies $L_\beta(q) > 0$. Therefore no nonzero element of $\mathbb{R}[C]_{\leq 2n}$ reduces the first core set, so $\mathcal{CV}^{(1)}(\beta) = C$. The defining family is then unchanged at every subsequent iteration, and hence $\mathcal{CV}(\beta) = C$.

It remains to consider the case in which $M_2^{(n)}$ is singular. There then exists $v \in \mathbb{R}[x, y]_{\leq n-1}$, nonzero modulo (P_1) , such that $L_\beta(\chi_2 P_2 v^2) = 0$. Since $\chi_2 P_2$ is strictly positive on C_1 and vanishes on C_2 , we have $\text{supp } \mu \subseteq C_2 \cup S$, where $S := C_1 \cap \mathcal{Z}_\mathbb{R}(v)$. The restriction $v|_{C_1}$ is a nonzero univariate polynomial of degree at most $n-1$, and therefore $\text{card } S \leq n-1$. Lemma 8 gives $3n \leq (2n+1) + \text{card } S \leq 3n$, so $\text{card } S = n-1$.

Write $\mu = \mu_1 + \mu_2$, where $\text{supp } \mu_i \subseteq C_i$. This decomposition is unambiguous because $C_1 \cap C_2 = \emptyset$. Moreover,

$$3n \leq \text{rank } M(n)[\mu_2] + \text{card}(\text{supp } \mu_1) \leq (2n+1) + \text{card } S = 3n.$$

Thus $\text{rank } M(n)[\mu_2] = 2n+1$ and $\text{supp } \mu_1 = S$. In particular, every point of S has positive μ_1 -mass, and $M(n)[\mu_2]$ is P_2 -pure. By Theorem 10, the core variety of the moment sequence generated by μ_2 is C_2 .

For every $z \in C_2$, there is therefore a representing measure $\nu_{2,z}$ for the moments of μ_2 whose support contains z . The measure $\mu_1 + \nu_{2,z}$ represents β , so $z \in \mathcal{CV}(\beta)$. Since every point of S belongs to $\text{supp } \mu$, we conclude that $\mathcal{CV}(\beta) = C_2 \cup S$, where $\text{card } S = n-1$. $\square$

It remains to consider reducible cubics for which the line and the irreducible conic have no common non-real affine point. The following theorem treats all possible real affine intersection configurations simultaneously.

Theorem 28. *Let $n \geq 3$, let $P = P_1 P_2 \in \mathbb{R}[x, y]$, where $\deg P_1 = 1$, $\deg P_2 = 2$, and P_2 is irreducible with infinite real zero set. Set*

$$C_i := \mathcal{Z}_{\mathbb{R}}(P_i), \quad C := C_1 \cup C_2,$$

and assume that every common affine zero of P_1 and P_2 is real, that is,

$$\mathcal{Z}_{\mathbb{C}}(P_1) \cap \mathcal{Z}_{\mathbb{C}}(P_2) \subseteq \mathbb{R}^2.$$

Let $r := \text{card}(C_1 \cap C_2)$, and let $M(n)(\beta)$ be a positive semidefinite moment matrix that is P -pure. Then exactly one of the following alternatives holds:

- (i) $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = C$, and a representing measure exists;
- (ii) $r \leq 1$ and

$$\mathcal{CV}(\beta) = C_1 \cup S_2,$$

where $S_2 \subseteq C_2 \setminus C_1$ and $\text{card } S_2 = 2n - 1$; a representing measure exists;

- (iii) $r \leq 1$ and

$$\mathcal{CV}(\beta) = C_2 \cup S_1,$$

where $S_1 \subseteq C_1 \setminus C_2$ and $\text{card } S_1 = n - 1$; a representing measure exists;

- (iv) $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n)(\beta) = 3n$, and there exists a unique representing measure whose support is $\mathcal{CV}(\beta)$;
- (v) $\mathcal{CV}(\beta) = \emptyset$, and no representing measure exists.

In particular, if $r = 2$, only alternatives (i), (iv), and (v) can occur.

Proof. By P -purity, $\mathcal{V}(\beta) = C$ and $\text{rank } M(n)(\beta) = \dim \mathcal{P}_n - \dim \mathcal{P}_{n-3} = 3n$.

If $\mathcal{CV}(\beta) = \emptyset$, the Core Variety Theorem gives alternative (v). Hence suppose that $\mathcal{CV}(\beta) \neq \emptyset$. By the same theorem, β has a C -representing measure.

By Lemma 18(i), there exists $F \in \mathbb{R}[x, y]_{\leq 2n}$ such that $F|_C \geq 0$, $L_{\beta}(F) = 0$, and

$$W := \mathcal{CV}^{(1)}(\beta) = C \cap \mathcal{Z}_{\mathbb{R}}(F).$$

We distinguish four cases according to which factors of P divide F .

If $P_1 P_2 \mid F$, then $W = C$. Since $\mathcal{CV}^{(1)}(\beta) = \mathcal{CV}^{(0)}(\beta)$, the defining families at all subsequent iterations coincide. Thus $\mathcal{CV}(\beta) = C$, giving alternative (i).

Suppose that $P_1 \nmid F$ and $P_2 \nmid F$. The restriction of F to each component is nonzero and nonnegative. By Lemma 19 and Bézout's theorem, F has at most n distinct zeros on C_1 and at most $2n$ distinct zeros on C_2 . Hence $\text{card } W \leq 3n$, and

$$3n \leq \text{card } \mathcal{CV}(\beta) \leq \text{card } W \leq 3n.$$

Therefore $\text{card } \mathcal{CV}(\beta) = 3n$, and [13, Theorem 2.20] gives the unique representing measure supported on $\mathcal{CV}(\beta)$. This is alternative (iv).

Suppose next that $P_1 \mid F$ and $P_2 \nmid F$. Write $F = P_1 H$, where $\deg H \leq 2n - 1$, and set

$$S_2 := (C_2 \setminus C_1) \cap \mathcal{Z}_{\mathbb{R}}(H).$$

Then $W = C_1 \cup S_2$. At every point of S_2 , the factor P_1 is locally nonzero. Since $P_1 H$ is nonnegative on C_2 , every zero of $H|_{C_2}$ in S_2 has even local intersection multiplicity. Thus

$2 \text{ card } S_2 \leq 2 \deg H \leq 4n - 2$, so $\text{card } S_2 \leq 2n - 1$. Since $3n - \rho_n(C_1) = 2n - 1$, Lemma 27 gives

$$\mathcal{CV}(\beta) = C_1 \cup S_2, \quad \text{card } S_2 = 2n - 1.$$

If $r = 2$, let R, Q be the two points of $C_1 \cap C_2$. Bézout's theorem implies that both intersections are transverse. Since $P_1 H$ is nonnegative on C_2 , $H|_{C_2}$ has odd vanishing order at R and Q . Consequently,

$$2 + 2 \text{ card } S_2 \leq 2 \deg H \leq 4n - 2,$$

which gives $\text{card } S_2 \leq 2n - 2$, contradicting $\text{card } S_2 = 2n - 1$. Therefore this case can occur only when $r \leq 1$, and it gives alternative (ii).

Finally, suppose that $P_2 \mid F$ and $P_1 \nmid F$. Write $F = P_2 H$, where $\deg H \leq 2n - 2$, and set

$$S_1 := (C_1 \setminus C_2) \cap \mathcal{Z}_{\mathbb{R}}(H).$$

Then $W = C_2 \cup S_1$. Every zero of $H|_{C_1}$ in S_1 has even multiplicity, so $2 \text{ card } S_1 \leq \deg H \leq 2n - 2$. Hence $\text{card } S_1 \leq n - 1$. Since $3n - \rho_n(C_2) = n - 1$, Lemma 27 gives

$$\mathcal{CV}(\beta) = C_2 \cup S_1, \quad \text{card } S_1 = n - 1.$$

If $r = 2$, the polynomial $P_2|_{C_1}$ has simple zeros at the two points R, Q . Nonnegativity of $P_2 H$ on C_1 forces $H|_{C_1}$ to have odd vanishing order at both points. Therefore

$$2 + 2 \text{ card } S_1 \leq \deg(H|_{C_1}) \leq 2n - 2,$$

so $\text{card } S_1 \leq n - 2$, contradicting $\text{card } S_1 = n - 1$. Thus this case can also occur only when $r \leq 1$, and it gives alternative (iii).

The four divisibility cases exhaust all possibilities and complete the proof. $\square$

The next example realizes alternative (ii): for $n = 3$, the core variety consists of the entire line component together with exactly $2n - 1 = 5$ points on the conic component.

Example 29. Let $Q(x, y) := x^2 + y^2 - 2y$, $P := yQ$, $C_\ell := \mathcal{Z}_{\mathbb{R}}(y)$, and $C_c := \mathcal{Z}_{\mathbb{R}}(Q)$. These two components meet only at $o := (0, 0)$. Set $S_\ell := \{(\pm 1, 0), (\pm 2, 0)\}$ and $S_c := \{(\pm 1, 1), (0, 2), (\pm \frac{\sqrt{3}}{2}, \frac{1}{2})\}$, and let $\mu_0 := \sum_{z \in S_\ell \cup S_c} \delta_z$. Denote by $\beta^{(6)}$ the truncated moment sequence generated by μ_0 .

The moment matrix $M(3)$ is positive semidefinite, P -pure, and has rank 9. Indeed, suppose that $f \in \mathbb{R}[x, y]_{\leq 3}$ vanishes on $S_\ell \cup S_c$. Since $f(x, 0)$ has the four distinct zeros $\pm 1, \pm 2$, it vanishes identically, and hence $f = yh$ for some $h \in \mathbb{R}[x, y]_{\leq 2}$. Every point of S_c has a nonzero y -coordinate, so h vanishes at five distinct points of the irreducible conic C_c . Bézout's theorem therefore implies that h is a scalar multiple of Q . Thus every cubic relation is a scalar multiple of P .

Write $\mu_0 = \mu_\ell + \mu_c$, where $\mu_\ell := \sum_{z \in S_\ell} \delta_z$ and $\mu_c := \sum_{z \in S_c} \delta_z$. The four evaluation vectors associated with S_ℓ are linearly independent in degree three. Thus $M(3)[\mu_\ell]$ has rank 4 = $\rho_3(C_\ell)$ and is y -pure. By Theorem 10, its core variety is C_ℓ . Keeping μ_c fixed and varying a representing measure for the moments of μ_ℓ therefore shows that $C_\ell \subseteq \mathcal{CV}(\beta^{(6)})$. Moreover, $S_c \subseteq \text{supp } \mu_0 \subseteq \mathcal{CV}(\beta^{(6)})$.

For the reverse inclusion, set

$$F(x, y) := y(2 - y)(y - 1)^2(2y - 1)^2.$$

This polynomial has degree 6, vanishes on C_ℓ , and is nonnegative on C_c , because $0 \leq y \leq 2$ on the circle C_c . Its zero set on C is precisely $C_\ell \cup S_c$, and $L_\beta(F) = 0$. Hence $\mathcal{CV}(\beta^{(6)}) \subseteq \mathcal{CV}^{(1)}(\beta^{(6)}) \subseteq C_\ell \cup S_c$, proving that

$$\mathcal{CV}(\beta^{(6)}) = C_\ell \cup S_c.$$

This realizes alternative (ii) of Theorem 28.

The following example realizes alternative (iii): for $n = 3$, the core variety consists of the entire conic component together with exactly $n - 1 = 2$ points on the line component.

Example 30. Let $Q(x, y) := x^2 + y^2 - 2y$, $P := yQ$, $C_\ell := \mathcal{Z}_\mathbb{R}(y)$, and $C_c := \mathcal{Z}_\mathbb{R}(Q)$. Set $S_\ell := \{(\pm 1, 0)\}$, and let S_c consist of the seven points $(\pm 1, 1)$, $(0, 2)$, $(\pm \frac{\sqrt{3}}{2}, \frac{1}{2})$, and $(\pm \frac{15}{16}, \frac{16 - \sqrt{31}}{16})$. Define $\mu_0 := \sum_{z \in S_\ell \cup S_c} \delta_z$, and let $\beta^{(6)}$ be its truncated moment sequence.

The matrix $M(3)$ is positive semidefinite, P -pure, and has rank 9. To see this, suppose that $f \in \mathbb{R}[x, y]_{\leq 3}$ vanishes on $S_\ell \cup S_c$. Since f vanishes at seven distinct points of the irreducible conic C_c , Bézout's theorem implies that $f = Q\ell$ for some linear polynomial ℓ . Moreover, $Q(\pm 1, 0) = 1$, so $\ell(\pm 1, 0) = 0$. It follows that ℓ is a scalar multiple of y , and hence f is a scalar multiple of P .

Write $\mu_0 = \mu_\ell + \mu_c$ according to the two displayed components. The same Bézout argument shows that evaluation from $\mathbb{R}[C_c]_{\leq 3}$ onto $\mathbb{R}^{S_c}$ is an isomorphism. Hence $M(3)[\mu_c]$ has rank $7 = \rho_3(C_c)$ and is Q -pure. Theorem 10, applied while keeping μ_ℓ fixed, gives $C_c \subseteq \mathcal{CV}(\beta^{(6)})$. Also, $S_\ell \subseteq \text{supp } \mu_0 \subseteq \mathcal{CV}(\beta^{(6)})$.

Conversely, the degree-six polynomial

$$F(x, y) := Q(x, y)(x^2 - 1)^2$$

vanishes on C_c , while $F(x, 0) = x^2(x^2 - 1)^2 \geq 0$ on C_ℓ . It vanishes on $\text{supp } \mu_0$, so $L_\beta(F) = 0$. Since

$$C \cap \mathcal{Z}_\mathbb{R}(F) = C_c \cup \{(\pm 1, 0)\},$$

the definition of the first core iteration yields

$$\mathcal{CV}(\beta^{(6)}) = C_c \cup \{(\pm 1, 0)\}.$$

This realizes alternative (iii) of Theorem 28.

6. THE CORE VARIETY $\mathcal{CV}(\beta)$ FOR MOMENT MATRICES $M(n)$ WITH A REDUCIBLE CUBIC RELATION CONSISTING OF THREE LINES

In this section, we characterize the possible cases for the core variety when the moment matrix $M(n)$ admits a reducible cubic column relation consisting of three lines.

Theorem 31. *Let $n \geq 3$, and let $P = P_1 P_2 P_3 \in \mathbb{R}[x, y]$ be a reducible cubic polynomial consisting of three distinct real linear factors. Let $M(n) \equiv M(n)(\beta)$ be a positive*

semidefinite moment matrix that is P -pure. Then exactly one of the following alternatives holds:

- (i) $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = \mathcal{Z}_{\mathbb{R}}(P)$, and a representing measure exists;
- (ii) For some permutation (i_1, i_2, i_3) of $(1, 2, 3)$, $\mathcal{CV}(\beta) = \mathcal{Z}_{\mathbb{R}}(P_{i_1}) \cup \mathcal{Z}_{\mathbb{R}}(P_{i_2}) \cup S$, where $S \subseteq \mathcal{Z}_{\mathbb{R}}(P_{i_3}) \setminus (\mathcal{Z}_{\mathbb{R}}(P_{i_1}) \cup \mathcal{Z}_{\mathbb{R}}(P_{i_2}))$ and $\text{card } S = n - 1$; a representing measure exists.
- (iii) $\text{card } \mathcal{CV}(\beta) = \text{rank } M(n)(\beta) = 3n$, and there exists a unique representing measure whose support coincides with $\mathcal{CV}(\beta)$;
- (iv) $\mathcal{CV}(\beta) = \emptyset$, and no representing measure exists.

Proof. Set $C_i := \mathcal{Z}_{\mathbb{R}}(P_i)$ and $C := C_1 \cup C_2 \cup C_3$. Since $M(n)(\beta)$ is P -pure, $\mathcal{V}(\beta) = C$ and

$$\text{rank } M(n)(\beta) = \dim \mathcal{P}_n - \dim \mathcal{P}_{n-3} = 3n.$$

If $\mathcal{CV}(\beta) = \emptyset$, the Core Variety Theorem shows that no representing measure exists, giving alternative (iv). Hence suppose that $\mathcal{CV}(\beta) \neq \emptyset$, and choose a finitely atomic representing measure μ with $\text{supp } \mu \subseteq \mathcal{CV}(\beta)$.

By Lemma 18(i), there exists $F \in \mathbb{R}[x, y]_{\leq 2n}$ such that $F|_C \geq 0$, $L_\beta(F) = 0$, and

$$W := \mathcal{CV}^{(1)}(\beta) = C \cap \mathcal{Z}_{\mathbb{R}}(F).$$

We distinguish four cases according to the number of distinct factors of P dividing F .

Case 1: $P_1 P_2 P_3 \mid F$. Then $W = C$. Since $\mathcal{CV}^{(1)}(\beta) = \mathcal{CV}^{(0)}(\beta) = C$, the families defining the first and second core iterations coincide. Inductively, $\mathcal{CV}^{(j)}(\beta) = C$ for every $j \geq 1$. Thus $\mathcal{CV}(\beta) = C$, giving alternative (i).

Case 2: none of P_1, P_2, P_3 divides F . For every i , the restriction $F|_{C_i}$ is nonzero and nonnegative. Lemma 19 and Bézout's theorem therefore give

$$\text{card}(C_i \cap \mathcal{Z}_{\mathbb{R}}(F)) \leq n.$$

Hence $\text{card } W \leq 3n$. Since $\text{supp } \mu \subseteq \mathcal{CV}(\beta) \subseteq W$,

$$3n = \text{rank } M(n)(\beta) \leq \text{card}(\text{supp } \mu) \leq \text{card } W \leq 3n.$$

Thus $\mathcal{CV}(\beta) = W$, $\text{card } W = 3n$, and the extremal uniqueness theorem gives a unique representing measure whose support is W . This is alternative (iii).

Case 3: exactly one factor divides F . After relabeling, write $F = P_i H$, where $\deg H \leq 2n - 1$ and $P_j \nmid H$, $P_k \nmid H$, with $\{i, j, k\} = \{1, 2, 3\}$. For $m \in \{j, k\}$, set

$$S_m := (C_m \setminus C_i) \cap \mathcal{Z}_{\mathbb{R}}(H).$$

Then $W = C_i \cup S_j \cup S_k$.

At every $z \in S_m$, the factor P_i is locally nonzero. Therefore

$$I_z(P_m, H) = I_z(P_m, F) \in 2\mathbb{N}$$

by Lemma 19. Since $P_m \nmid H$, Bézout's theorem gives

$$2 \text{card } S_m \leq \deg H \leq 2n - 1,$$

and hence $\text{card } S_m \leq n - 1$. Applying Lemma 8 to μ , we obtain

$$3n \leq \rho_n(C_i) + \text{card}(S_j \cup S_k) \leq (n + 1) + 2(n - 1) = 3n - 1,$$

a contradiction. Thus this case cannot occur when the core is nonempty.

Case 4: exactly two factors divide F . After relabeling, write $F = P_i P_j H$, where $\deg H \leq 2n - 2$ and $P_k \nmid H$. Put

$$D := C_i \cup C_j, \quad S := (C_k \setminus D) \cap \mathcal{Z}_{\mathbb{R}}(H).$$

Then $W = D \cup S$. At every $z \in S$, the factor $P_i P_j$ is locally nonzero, so

$$I_z(P_k, H) = I_z(P_k, F) \in 2\mathbb{N}.$$

Consequently, Bézout's theorem gives

$$2 \text{ card } S \leq \deg H \leq 2n - 2,$$

and therefore $\text{card } S \leq n - 1$. Since $\rho_n(D) = 2n + 1$, we have

$$\rho_n(D) + \text{card } S \leq (2n + 1) + (n - 1) = 3n.$$

Lemma 27 now yields

$$\mathcal{CV}(\beta) = D \cup S, \quad \text{card } S = n - 1.$$

This is alternative (ii).

The four cases are mutually exclusive and exhaust all possible divisibility patterns for F , completing the proof. $\square$

Remark 32. Alternative (ii) can occur only when the three lines are concurrent or when they are pairwise parallel.

Proof. Assume that alternative (ii) occurs, and use the notation from Case 4 of the proof: $F = P_i P_j H$, $D = C_i \cup C_j$, and $S \subseteq C_k \setminus D$, with $\deg H \leq 2n - 2$ and $\text{card } S = n - 1$. If the three lines are neither concurrent nor pairwise parallel, then C_k meets D at a point z at which exactly one of P_i, P_j vanishes. Hence $P_i P_j|_{C_k}$ has a simple zero at z . Because $F|_{C_k} \geq 0$, the restriction $H|_{C_k}$ has odd vanishing order at z . At each point of S , it has even vanishing order. Therefore $1 + 2 \text{ card } S \leq \deg(H|_{C_k}) \leq 2n - 2$, whereas $1 + 2 \text{ card } S = 2n - 1$, a contradiction. $\square$

7. POSITIVE DEFINITE CASES

We conclude with two observations about low-order positive definite moment sequences.

Recall that a positive definite $M(1)$ has infinitely many three-atomic representing measures.

Proposition 33. *If $M(1) \equiv M(1)(\beta)$ is positive definite and $\beta_{00} > 0$, then $\mathcal{CV}(\beta) = \mathcal{V}(\beta)$.*

Proof. By invariance under invertible affine changes of variables and positive scaling, we may assume that

$$M(1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(see [11]). If a quadratic polynomial p lies in $\ker L_\beta$, then

$$p(x, y) = ax^2 + bxy + cy^2 + dx + ey - a - c,$$

where $a, b, c, d, e \in \mathbb{R}$. If $p \geq 0$ on $\mathbb{R}^2$, then $a, c \geq 0$, while $p(0, 0) = -a - c \geq 0$. Hence $a = c = 0$; nonnegativity then forces $b = d = e = 0$. Thus the zero polynomial is the only nonnegative quadratic annihilated by L_β , so $\mathcal{CV}^{(1)}(\beta) = \mathcal{CV}^{(0)}(\beta) = \mathbb{R}^2$. The defining family does not change at later iterations, and therefore $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = \mathbb{R}^2$. $\square$

Proposition 34. *Assume that the principal submatrix $M(1)$ of $M(2)$ is positive definite. If $M(2) \succeq 0$ and $\text{rank } M(2) \geq 5$, then $\mathcal{CV}(\beta) = \mathcal{V}(\beta)$.*

Proof. If $\text{rank } M(2) = 5$, the kernel is generated by a single quadratic column relation, so the assertion follows from Theorem 10. If $\text{rank } M(2) = 6$, then $M(2)$ is positive definite, and Proposition 6 gives $\mathcal{CV}(\beta) = \mathcal{V}(\beta) = \mathbb{R}^2$. $\square$

Remark 35. Even when the core variety is infinite, a prescribed point need not belong to the support of a minimal representing measure. For example, consider

$$M(1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

If a representing measure contains $(0, 0)$ with mass $u > 0$, then subtracting that atom gives

$$M(1) = \begin{bmatrix} 1-u & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + uP_1(0, 0) = \begin{bmatrix} 1-u & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} u & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

If the original measure had three atoms, the residual matrix would have a representing measure with at most two atoms and therefore rank at most two. This forces $u = 1$. But the resulting residual matrix $\text{diag}(0, 1, 1)$ is not recursively generated: it has the column relation $1(X, Y) = \mathbf{0}$, whereas its X - and Y -columns are nonzero. Hence it has no representing measure. Every representing measure for $M(1)$ whose support contains $(0, 0)$ must therefore have at least four atoms, although $M(1)$ itself has three-atomic representing measures.

Remark 36. Let D be either a line or an irreducible conic with infinite real zero set, and let a representing measure decompose as $\mu = \mu_D + \mu'$, where μ_D is supported on D . If $\text{rank } M(n)[\mu_D] = \rho_n(D)$, then $M(n)[\mu_D]$ is pure with respect to a defining polynomial of D . By Theorem 10, its core variety is D , so varying a representing measure for this component while keeping μ' fixed gives infinitely many representing measures for β . In particular, the rank hypothesis holds if $\text{supp } \mu_D$ contains $n + 1$ distinct points of a line, or $2n + 1$ distinct points of an irreducible conic with infinite real locus, because the corresponding evaluation vectors are linearly independent.

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